Hubbard dynamics on a fixed qubit array: protection, mitigation, and achievable lattice size

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Abstract

We compare three uses of the same nearest-neighbor array: physical-qubit computation with error mitigation, partially protected STAR computation, and fully fault-tolerant surface-code computation. The application is a local doublon observable after a Fermi–Hubbard quench. We organize the comparison around the largest lattice accessible with a given number of physical qubits and one week of execution. A common accounting model separates sampling, mitigation exposure, logical protection, non-Clifford supply, and parallel replicas. Explicit circuit compilation and small-patch dynamics identify the mechanisms that control the comparison: idle exposure depends strongly on the schedule; STAR depends on both total rotation angle and accumulated injection floors; FT depends on supply and consumption together; and Trotter refinement changes the cost of these architectures differently. The compiled four-step reference curves are conditional circuit comparisons, since that step count fails the observable-accuracy target on both square patches tested. We retain this distinction while developing a common basis for architecture optimization and hybrid strategies. For immediate discussion, we also provide a simpler analytical comparison with experimentally anchored circuit scales, measured finite-patch time dependence, and per-arm error-budget optimization. Direct 4×4 checks validate the Trotter allowances at the selected headline step counts; time extensions assess ordering-dependent finite-size corrections. Layout and source-supply sensitivities show why faster consumption need not translate into proportional area savings. We now vary physical error and precision with the calibrated workload held fixed, and test local encoding and repeated-angle synthesis as separate algorithm choices. Hamming-weight phasing reduces the modeled strict-precision FT area by factors of approximately six at the three sizes tested, while its workspace can outweigh the benefit at loose precision.

1 Physical error, precision, and algorithm choice

We lead the workshop example with 0.01 absolute error on the local doublon observable and retain 0.001 as a second principal target. Figure 1 varies p from 10^{-3} to 10^{-4} while keeping the clocks, Hamiltonian, initial state, observable, and finite-patch Trotter model fixed. The $p = 10^{-3}$ central rows reproduce all 144 corresponding entries of the previous added-patch comparison. The new sweep contains 864 scenario points, with per-arm error allocation and source counts reoptimized. The model remains a finite-lattice workload comparison: at fixed time, an independently justified local-patch approximation could change what increasing L means.

Source assumptions remain visible. We do not infer a new cultivation protocol from a change in p. The held scenario keeps $q_T = 2 \times 10^{-9}$, a 1024-qubit source, 100 preparation cycles, and the distance-15 handoff constraint. A second scenario assumes $q_T(p) = 2 \times 10^{-9}(p/10^{-3})^3$ and an injection floor $f(p) = 10^{-5}(p/10^{-3})$, with source area and cycles unchanged. A third keeps both error floors fixed and multiplies source preparation cycles by four. These choices bracket specified sensitivities; they are not calibrated lower-p implementations or guaranteed bounds on an optimal source. In

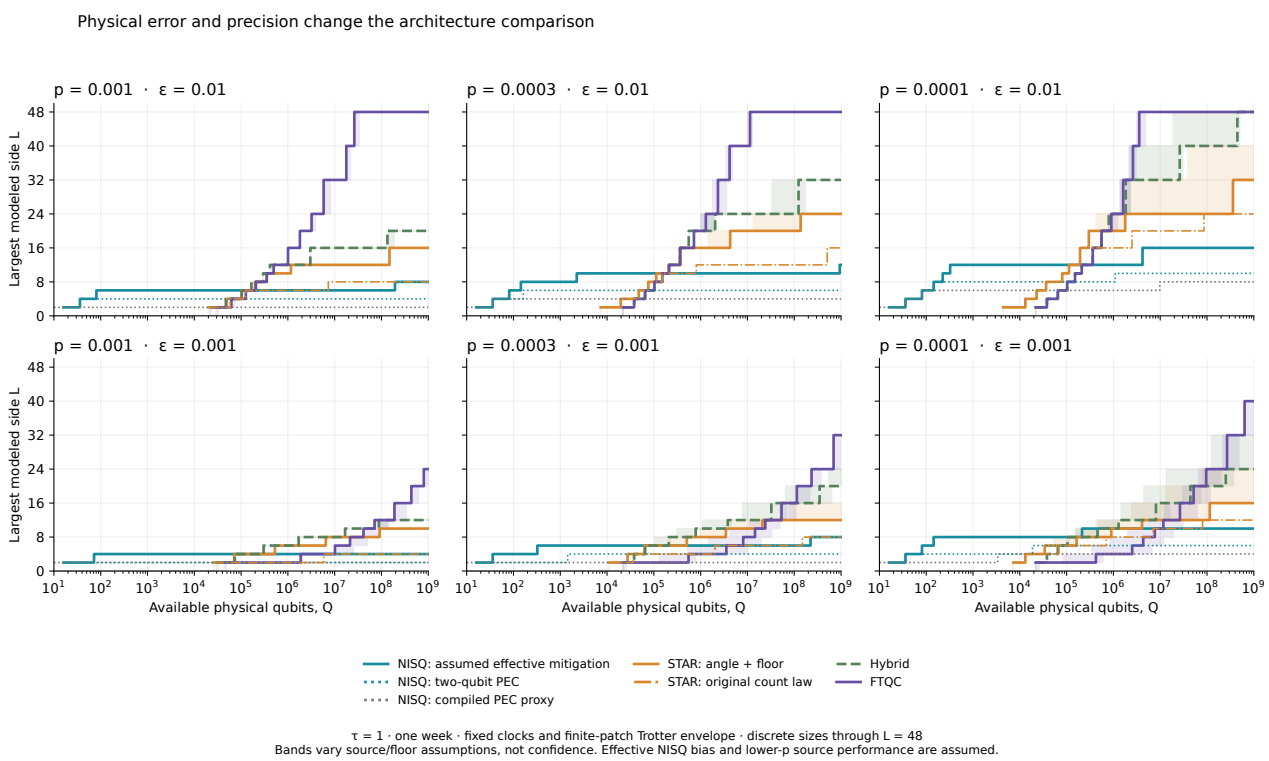


Figure 1: Achievable lattice side versus available physical qubits at $\tau = 1$ and a one-week cutoff. Columns vary physical error; rows show 0.01 and 0.001 absolute error. Solid encoded curves retain the source and injection floor at their $p = 10^{-3}$ reference values. Bands span the explicit source/floor scenarios described in the text, not statistical confidence intervals. The count-law STAR curve changes the injection exposure while retaining the same delivery model. NISQ retains separate assumed-effective, two-qubit PEC, and compiled-PEC scenarios. Only the sampled sizes are shown; $L = 48$ is the sweep ceiling. Larger-lattice accuracy transfer and effective mitigation bias remain assumptions.

particular, the inherited handoff convention imposes $d \geq 15$ on the FT/hybrid data block as well, so its low-p area floor is a model choice. The ideal cleanup term $35q_T^3$ accounts for input-state errors. Faults in the cleanup block and waiting storage are charged separately through the protected tile-time volume and logical-error budget.

At $Q = 10^6$ and $\epsilon = 0.01$, lowering p from 10^{-3} to 10^{-4} raises the held-scenario STAR capacity from 10×10 to 20×20, and FT/hybrid capacity from 12×12 to 24×24. The NISQ effective and two-qubit PEC capacities increase from sides 6 and 4 to 12 and 8, respectively. At $\epsilon = 0.001$, the corresponding STAR and FT sides are 6 and 2 at $p = 10^{-3}$, and 10 and 4 at $p = 10^{-4}$. The latter comparison is sensitive to the magic-state assumption: for the strict 4×4 FT task at $p = 10^{-4}$, holding q_T fixed costs 4.23×10^5 qubits, whereas the assumed cubic improvement gives 8.47×10^4. Lower physical error benefits every arm, but the relative window cannot be specified by p without the source assumptions.

Injection law as a controlled switch. For the original injection model [1], we use the exact dephasing-inverse exposure

$$\chi_{\text{count}} = -2R\log(1 - 8p/15), \qquad \chi_{\text{angle}} = 4\alpha p\Theta + 4fR. \tag{1}$$

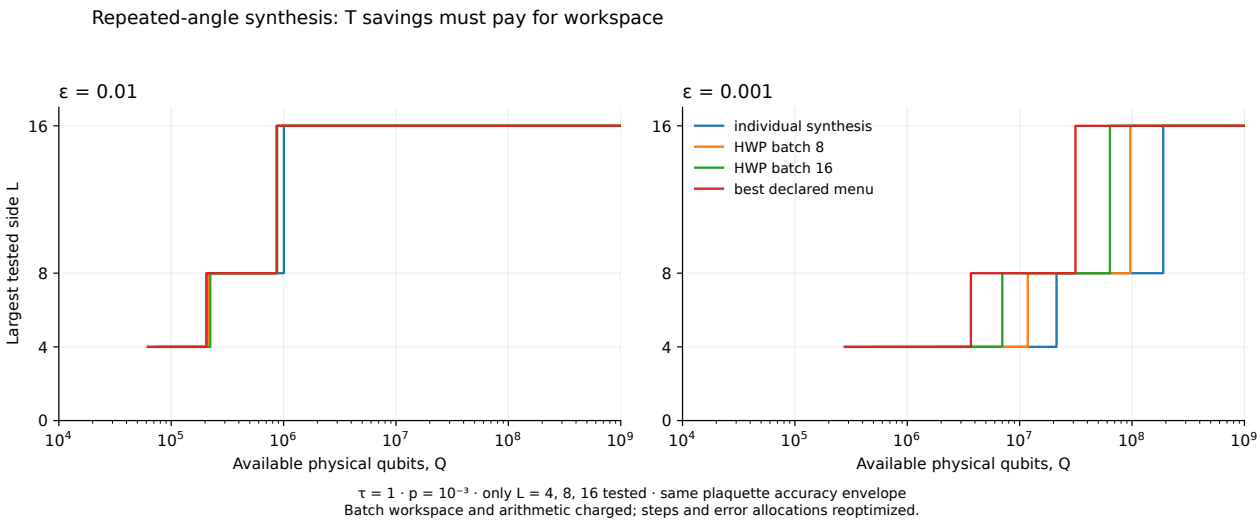


Figure 2: FT capacity with repeated-angle synthesis, for the three tested lattice sides 4, 8, and 16. The best-menu curve chooses individual synthesis or an allowed Hamming-weight batch, with steps and error budgets reoptimized. The circuit, plaquette accuracy envelope, source menu, and delivery convention are common. The staircase ends at the largest tested size; it is not an FT capacity plateau.

Both calculations use the same surface-code law and delivery timing, so this switch isolates the exposure law rather than reconstructing two complete physical protocols. At $p = 10^{-3}$, the optimized strict 4×4 STAR estimate changes from 7.14×10^4 qubits under the angle law to 5.87×10^6 under the count law. At $p = 10^{-4}$, both estimates fit in one 13,122-qubit copy. Thus a change in the injection law can dominate the strict comparison at the original noise point without producing a fixed architecture advantage at all p.

Repeated-angle synthesis changes the FT comparison. We implement Campbell's Hamming-weight phasing construction for equal-angle rotations within each plaquette stage [2]. A batch of b rotations uses

$$a_b = b - \mathrm{popcount}(b), \qquad k_b = \lfloor \log_2 b \rfloor + 1 \tag{2}$$

clean ancillas/measurement-assisted Toffolis and synthesized rotations, respectively. We charge four T states per Toffoli, synthesize all k_b weight-register rotations within the common error budget, and reuse the a_b ancillas between batches. The peak workspace is charged for the full shot; arithmetic adds two logical Clifford clocks per Toffoli to the reference timing model. We retain all fixed plaquette T gates and do not pool equal angles from different times. Batch size one exactly reproduces individual synthesis. The nearest-neighbor surgery layout and effective delivery ports remain uncompiled reference assumptions.

At $p = 10^{-3}$ and $\epsilon = 0.001$, batch size 16 reduces the 4×4 estimate from 1.88×10^6 to 2.78×10^5 qubits. The optimized circuits consume 192,332 and 80,123 T states per shot, with 33 and 36 steps, respectively; the latter uses 15 extra logical ancillas. The area improvements at $L = 8$ and 16 are factors 5.78 and 6.04, using batches of 64 and 63 live ancillas. At $\epsilon = 0.01$, individual synthesis remains cheaper at $L = 4$ and 8 because one copy already completes the campaign and the extra workspace raises its area. These results change the strict FT comparison substantially without changing the Hamiltonian simulation algorithm. The principal p sweep deliberately retains individual synthesis to preserve its frozen baseline.

At strict 4×4, neither selected circuit uses a cleanup stage. The lower T demand reduces the

magic-state error allocation from 38.5% to 16.0% and increases the statistical allocation from 29.8% to 58.0%. The reduced shot count explains why the area gain exceeds the reduction in T count: repeated-angle synthesis changes the optimal estimator budget as well as the cost of a circuit.

Local encoding and local ordering are different changes. The compact encoding maps hopping to Pauli operators of weight at most three and uses fewer than 1.5 qubits per fermionic mode on the square lattice [3]. We evaluate a $3N$-qubit register scenario, with four CNOTs per hopping Pauli and two per onsite Pauli, plus $8N$ assumed preparation gates. A two-CNOT hopping variant assumes fusion; a routing variant doubles movement and placement costs. Preparation, embedding of both spin sectors, and gate utilization are explicitly uncompiled assumptions. The generic parity gadgets use 6,754 two-qubit gates at $L = 4, r = 8$, compared with approximately 3,455 in the native JW proxy. The fused hypothesis uses 3,682. Removing asymptotic routing growth therefore does not guarantee savings at the small lattices accessible under the present noise assumptions.

For STAR, the compact scenario enlarges the data register and separately tests four and sixteen assumed rotation lanes. Four lanes cost more area than the JW reference at $L = 4$, and sixteen lanes cost still more because the selected calculation already fits in one copy. Locality does not automatically pay for parallel delivery. These resource scenarios retain the existing termwise error envelope; transferring it to larger lattices remains an assumption.

A separate exact-sector diagnostic isolates ordering. Keeping the Hamiltonian, initial state, JW representation, and original onsite split fixed, a four-color bond permutation changes the $\tau = 1$ coefficient $|D_r - D|r^2$ from approximately 0.0344 to 0.0463 on 2×2 and from 0.0766 to 0.0913 on 3×3. Grouping all onsite terms instead gives 0.2231 and 0.2178, matching the plaquette results on these patches. These measurements show that a more parallel ordering need not reduce the local-observable error. They neither establish bulk convergence nor replace the frozen accuracy envelope.

2 Reconciliation of the workshop calculations

We evaluate the two supplied workshop implementations at 4×4, $\tau = 1, 2$, and 0.01 absolute error, with common elementary clocks and a common Gaussian doublon shot budget. The statistical allowance is 0.004 at $D_\star = 0.1$, giving 21,609 ideal shots. Taking 25% of the error budget for the larger of our two planning coefficients gives 11 and 23 steps. This step convention isolates resource differences; it is not an accuracy certificate for the different observables and unspecified circuits underlying the supplied models. We remove the IQUS physical-cone discount and explicitly fix its signal scale. The original implementations remain untouched, and their imported Python sources are preserved alongside the comparison data.

Matching these settings does not make the resource inventories identical. At $\tau = 1, r = 11$, the two-qubit gate counts are approximately 4,750, 3,520, and 2,640 for our native proxy, Martin, and IQUS, respectively. The corresponding STAR rotation counts are 2,443, 2,288, and 880. Their STAR shot times are approximately 50.3, 1.86, and 0.686 ms, reflecting different rotation parallelism and retry schedules even after the syndrome clock is matched. Native factory and logical-error conventions also remain different. These outputs diagnose assumptions rather than rank three fully specified machines.

An additional comparison applies every injection law to the identical ledger $R = 2{,}443$, $\Theta = 64$ at $p = 10^{-3}$. Martin's native attenuation convention gives $\chi = 0.9772$, the IQUS count convention gives 2.6042, the exact original-injection inverse gives 2.6066, and the angle-plus-floor law gives 0.4817. Thus even the phrase "count law" does not identify a common estimator convention. The differences do not collapse onto precision, evolution time, syndrome clock, and one injection switch alone: compilation, delivery, and the estimator must also be named. This shared-ledger calculation is an algebra check, not an empirical collapse test or independent evidence for a mitigation law.

3 Back-of-the-envelope+ discussion example

We use simple analytical laws to expose the architectural tradeoffs, with $p = 10^{-3}$, 10 ns gates, 100 ns measurement/reset, 240 ns syndrome cycles, $U/J = 4$, and a one-week budget B. The original reference below emphasizes $\epsilon = 0.001$; both precisions remain explicit in the current comparison of Section 1. We compare NISQ, STAR, FTQC, and the hybrid using a common total-error target. Each arm can choose its error allocation, step count, and resource-state supply within the declared search. This removes a fixed allocation that previously forced unnecessary cleanup at the strict 4×4 FT point. The current finite-patch capacity sensitivity is shown in Figure 9, and the directly checked 4×4 allocations in Table 1. We first develop the common model using the original 3×3 reference, retaining its longer-time and $\tau = L/4$ scenarios for comparison.

Ordering-specific time dependence. We extend exact fixed-sector calculations on 2×2 and 3×3 patches through $\tau = 8$, and check the two product formulas separately. The termwise evolution retains the ordering of the compiled Pauli circuit; the plaquette evolution retains its onsite/pink/gold splitting. Independent small-system comparisons check the factorized sector implementation against the qubit circuits and the sparse Hamiltonian evolution.

For each ordering $\mathcal{A}$ and measured time, we increase r until the last two estimates of $|D_r - D_{\text{exact}}|r^2$ agree within 5%. The observed r^{-2} dependence survives, but its coefficient is neither a single power of time nor independent of the ordering. On 3×3, the plaquette and termwise coefficients are respectively 0.218 and 0.0767 at $\tau = 1$, and 7.67 and 23.74 at $\tau = 8$. Figure 3 shows the full time dependence and patch sensitivity. At $\tau = 2$, the 4×3 plaquette and termwise coefficients are 1.136 and 0.702, compared with 0.926 and 0.381 on 3×3. At $\tau = 4$ they are 4.100 and 2.728, compared with 5.270 and 1.957. These differences preclude a claim of bulk convergence.

At the smaller step counts admitted by the optimizer, higher-order terms can matter. We therefore also measure $r = \lceil 8\tau \rceil$ and twice that value on 3×3. The coefficient used below is the largest sampled value across these finite-step checks and the asymptotic pair. This corrects an underestimate caught by direct validation at loose precision.

For the planning curves we use

$$C_{\mathcal{A}}^{\text{plan}}(\tau) = 1.1 \max_{s \le \tau,\ s \in \mathcal{T}} C_{\mathcal{A}}^{3\times 3}(s), \qquad \epsilon_T(r,\tau) = \frac{C_{\mathcal{A}}^{\text{plan}}(\tau)}{r^2}, \tag{3}$$

where $\mathcal{T}$ is the measured time grid. The running maximum avoids relying on isolated small-patch cancellations; it is a planning choice for the final-time observable, not a change to a whole-curve accuracy target. The 10% margin is a declared sensitivity, not a finite-size error bound. We retain pointwise-coefficient costs separately. No time power is fitted, and we do not extrapolate beyond $\tau = 8$. The 2×2 resource point uses its own measured coefficients; other sizes transfer the 3×3 reference. The $\tau = L/4$ panel therefore stops at $L = 32$. Transfer to larger lattices remains an assumption.

Joint error allocation. For each candidate we impose

$$\epsilon_{\text{stat}} = \epsilon - \epsilon_T - \epsilon_L - \epsilon_{\text{syn}} - \Gamma N_T q_T - \epsilon_{\text{res}} > 0. \tag{4}$$

Unused components vanish: termwise STAR has no synthesized rotations or supplied T states, and the hybrid synthesizes no variable rotations. The logical reserve satisfies $\Gamma V_L p_L(d) \le \epsilon_L$. Mitigated arms reserve $\epsilon_{\text{res}} = 0.1\epsilon$ for residual estimator bias, including channel mismatch and unmodeled SPAM. Its affordability is assumed. FT has no mitigation reserve; its modeled faults are charged to the logical and magic-state terms.

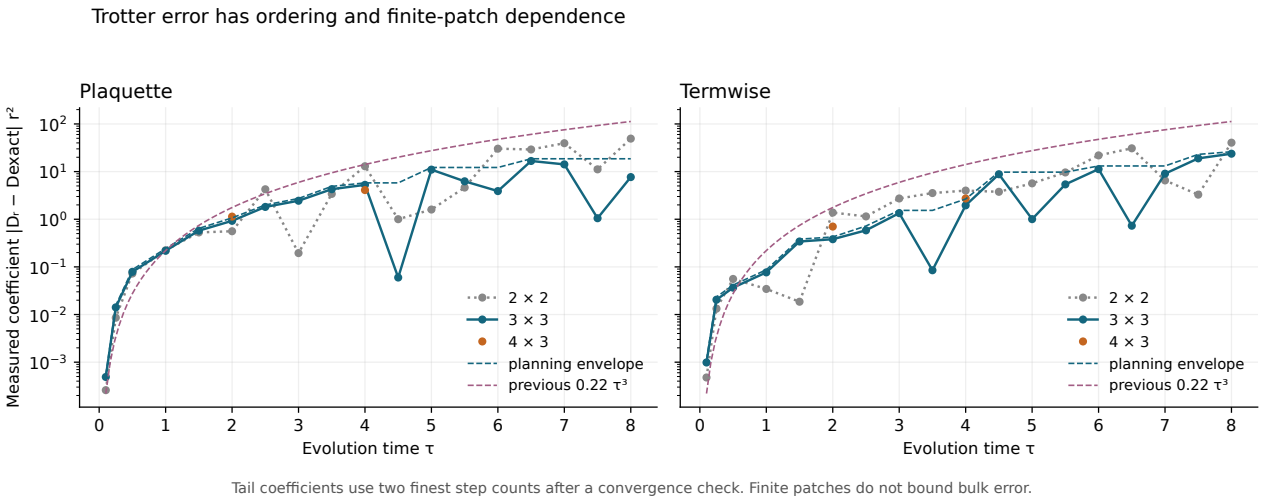


Figure 3: Local Trotter-error coefficients depend on time, ordering, and patch. Each point uses the finer pair after a step-convergence check. The dashed blue curve is the planning envelope in Eq. (3); the previous $0.22\tau^3$ prescription is shown for comparison. The patch differences are not a bulk error bound.

We search Trotter allowances, logical reserves, synthesis tolerances, and integer-step neighbors, with $r \geq \lceil 8\tau \rceil$ as a small-step guard. FT and hybrid independently compare raw cultivated states with one cleanup stage and optimize 1–16 sources. Both use the same source menu. The result is the best candidate in the declared search, not a global optimum over protocols or source-quality/throughput tradeoffs. The resulting per-arm allocations are recorded with each resource point.

For effective mitigation and encoded planning curves,

$$M_0 = \left\lceil \frac{1.96^2 D_*(1-D_*)}{\epsilon_{\text{stat}}^2} \right\rceil, \qquad Q_{\text{week}} = q \max\left(1, \left\lceil \frac{M_0 F t_{\text{shot}}}{B} \right\rceil\right), \qquad D_* = 0.1. \tag{5}$$

The PEC references retain the sufficient Hoeffding count from Eq. (14). Distance follows the common surface-code fit, with worst-case $\Gamma = \sqrt{F}$ amplification for mitigated logical and magic faults. The central delivery model has four effective parallel lanes; these are a throughput assumption, not a compiled dependency schedule.

The per-arm laws. NISQ uses the Willow circuit-count anchor [4],

$$G_2 = 4372 \frac{r}{3}\left(\frac{L}{6}\right)^3, \qquad D_{\text{circ}} = 294 \frac{r}{3}\frac{L}{6}, \qquad \log F_{\text{eff}} = \log(\eta_*)\frac{p}{10^{-3}}\frac{G_2}{4372}. \tag{6}$$

The central $\eta_* = 100$, with 10–1000 sensitivity, is an assumed sampling overhead, not a fit to experimental accuracy. Its bias reserve also remains uncalibrated. Two explicit PEC comparisons use two-qubit inverse-channel exposure, $\log F \simeq 4pG_2$, or the saved compiled full-noise exposure with bulk terms proportional to r and separate SPAM. The NISQ proxy transfers the plaquette error coefficient without asserting a match to the experimental ordering.

Termwise STAR uses

$$\Theta = (5L^2 - 4L)\tau, \qquad R = [16L(L-1) + 2L^2 - 2]r + 1, \qquad \log F_{\text{STAR}} = 4\alpha p\Theta + 4fR, \tag{7}$$

with $\alpha = 1.5$, $f = 10^{-5}$, and 18 code clocks per completed rotation [1, 5]. The scenario range varies f from zero to 10^{-4} and lanes from eight to one. Refinement leaves the angle term unchanged while

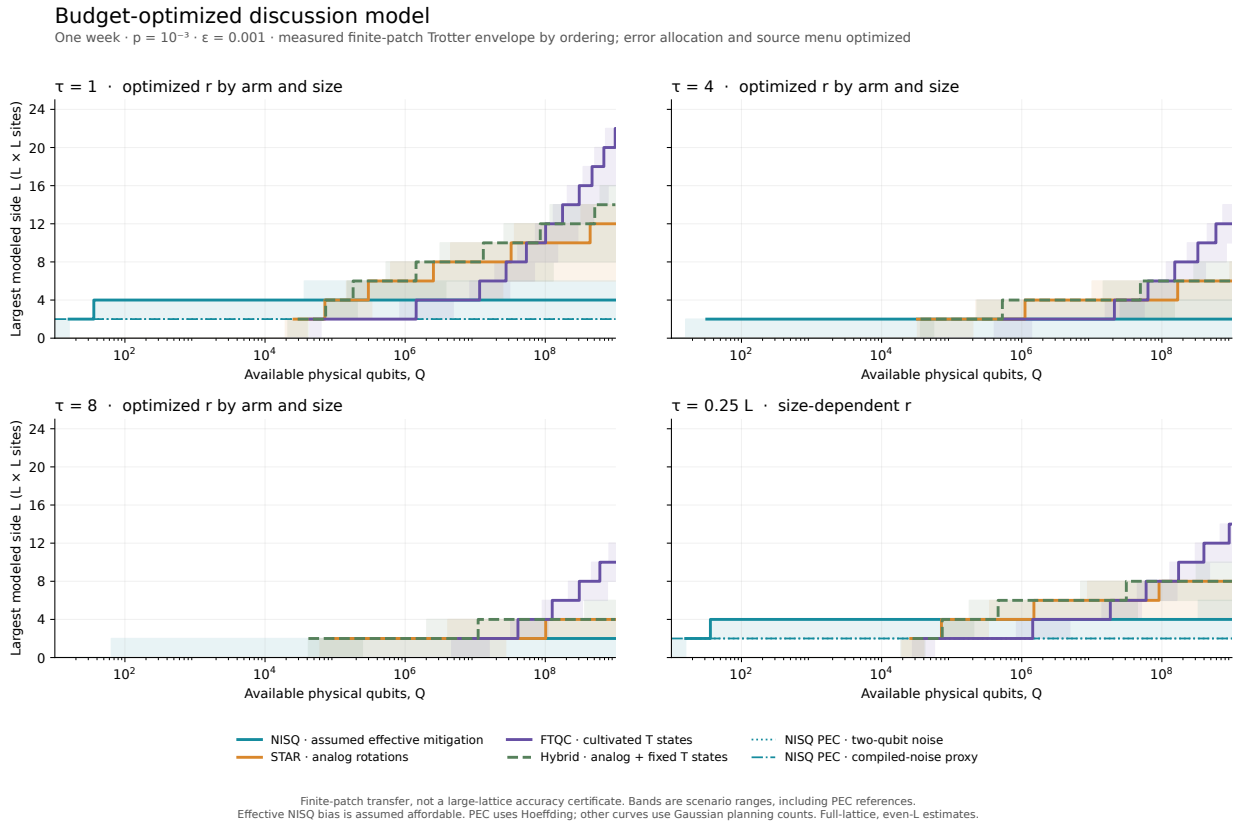


Figure 4: Budget-optimized discussion comparison at $\epsilon = 0.001$. Each arm selects its error allocation and step count using the measured finite-patch planning envelope. Lines are central assumptions; bands span delivery and mitigation scenarios, including NISQ PEC references. They do not represent uncertainty in bulk Trotter error. The fourth panel follows $\tau = L/4$. Full lattices and even L are modeled. Effective NISQ mitigation is an assumed estimator scenario, not a demonstrated accuracy result.

increasing the floor term and execution time. FT uses plaquette counts, logarithmic synthesis, and $F = 1$. The hybrid uses the same plaquette workload, injecting variable rotations and supplying fixed T states. It therefore has exponent $4\alpha p\Theta_{\mathrm{dyn}} + 4fR_{\mathrm{dyn}}$.

FT and hybrid share the rounded cultivation reference: 1024 source qubits, 100 syndrome cycles per state plus $2d$ handoff cycles, and raw state error 2×10^{-9} [6]. The optional 15-to-1 cleanup changes source error, footprint, and service together. Supply overlaps consumption; the reference charges nine code clocks per T and varies source service time by 0.5–4. These source models remain conditional implementations. Termwise STAR already avoids fixed T gates, so a hybrid advantage is algorithm-dependent.

Using the 3×3 planning reference at $\tau = 1$, the central areas are

Lattice	ϵ	NISQ effective	STAR	Hybrid	FTQC
4×4	.001	36	71442	73348	1.43×10^6
4×4	.01	36	46818	48420	61974
6×6	.001	3.64×10^{10}	298356	180798	1.16×10^7
6×6	.01	81	101802	103548	126268

For strict 4×4 FT, the selected $r = 31$ candidate allocates about 33.6% to statistics, 25.3% to Trotter error, 36.2% to magic-state error, 2% to logical faults, and 3% to synthesis. Cleanup is no longer

needed, reducing the previous 5.66 million-qubit estimate by about a factor of four. The strict 6×6 FT candidate still chooses cleanup. The hybrid improves on termwise STAR at strict 6×6, but not at 4×4. NISQ's effective estimate remains conditional on the specified sampling multiplier and affordable residual bias; its PEC alternatives are shown alongside it.

We separate the changes with an ablation that first optimizes the budget under the old cubic-time coefficient and then replaces that coefficient. A finer allocation grid improves the selected $\tau = 1$ costs by at most 18%. Direct checks at the optimized step counts fit the planning envelope on 3×3. At strict 4×4, $\tau = 8$, the optimized STAR floor exposure is 2.28, below its angle exposure 3.07; the earlier floor-dominated example depended on the assumed cubic time law. Neither validates the transfer from finite patches to large lattices. The earlier four-step compiled reference and fixed-allocation discussion results are retained as historical comparisons. The remainder develops their resource laws and implementation evidence.

4 Delivery rate and protected area

Faster consumption helps only when its extra area and state supply are included. We reoptimize the error budgets, source count, and replicas for three data-block designs from Litinski [7]. For n stored logical qubits their tile counts and worst-case T-consumption times are

$$(A, c_T)_{\text{compact}} = (1.5n + 3, 9), \quad (A, c_T)_{\text{intermediate}} = (2n + 4, 5), \quad (A, c_T)_{\text{fast}} = (2n + \sqrt{8n} + 1, 1). \tag{8}$$

Here c_T is in code clocks, each containing d syndrome cycles. We use $n = 2L^2 + 6$, round the tile count upward, and retain six additional tiles per delivery or injection lane. The fast-block expression is a reference area formula, not a placed floorplan. Its relative overhead at small L is larger than the asymptotic 4/3 ratio.

The earlier discussion model already assumes four effective lanes and divides its nine-clock T cost by four. It is therefore not the paper's serial compact block. For the whole-block comparison we do not divide the published T rate by the analog-lane count. We retain a separate, explicitly uncompiled scenario with four fast T ports, charging the fast data area and four lane allowances. Whether those ports can execute independent Pauli products at that rate requires a dependency and routing construction.

STAR retains eighteen clocks per analog rotation. Its effective parallelism is varied over $b_A \in \{1, 2, 4, 8, 16\}$, with area charged for every lane. The hybrid varies analog lanes and T consumption independently. For m sources and b_T T ports, the competing stages are

$$t_{\text{supply}} = \lceil N_T/m \rceil \, s(d) t_{\text{cyc}}, \tag{9}$$

$$t_{\text{consume}} = d \left[\frac{c_T N_T}{b_T} + \frac{18 R_{\text{analog}} + C_{\text{Cliff}}}{b_A} \right] t_{\text{cyc}}. \tag{10}$$

The shot duration uses their maximum, plus initialization and feedforward. The whole-block T rate sets $b_T = 1$. Both FT and hybrid receive the same source menu, extended through 1024 sources so faster consumers can be supplied; selected optima must lie below that endpoint. The menu search is discrete, and analog parallelism remains an assumption rather than a compiled schedule.

At strict 4×4, $\tau = 1$, the single fast block reduces FT's 1.43 million-qubit reference to 0.86 million. Four fast ports reduce it to 0.37 million, where state supply limits further speedup. Thus the reference does not receive a universal factor-of-nine improvement. STAR and hybrid select two analog lanes at this point, requiring about 61,000 and 63,000 qubits. At loose precision one replica often fits the week; then extra throughput can increase total area. Figures 5 and 6 separate these effects. These sensitivities retain the same 3×3 planning coefficients, so they isolate delivery from finite-size calibration.

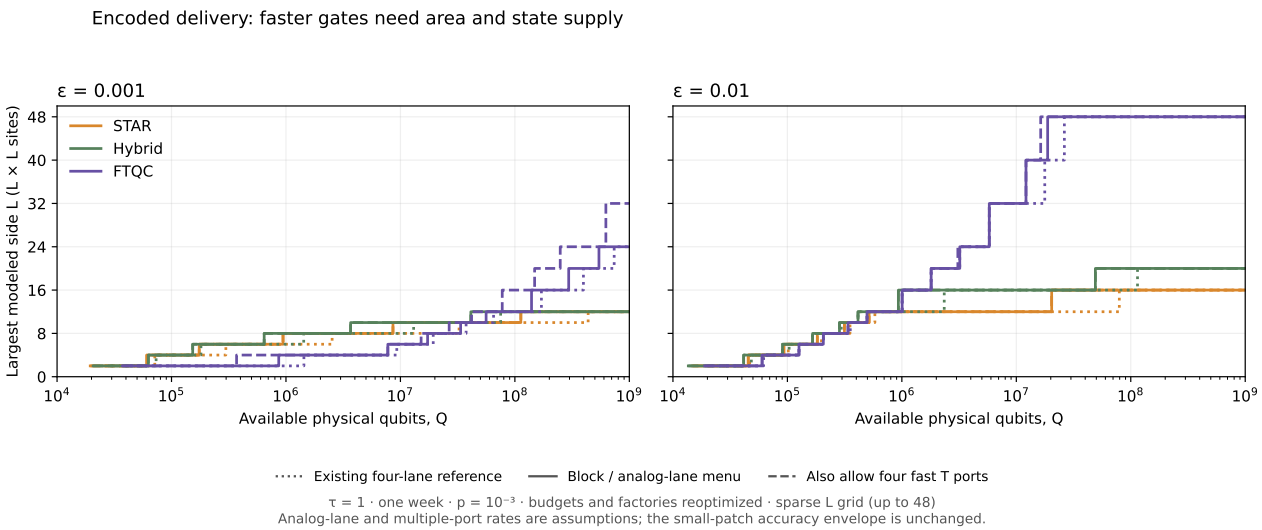


Figure 5: Encoded capacity at $\tau = 1$ after reoptimizing budgets and state supply. The dotted reference retains four effective lanes. The solid menu adds whole-block designs and analog-lane choices; the dashed menu also allows four fast T ports. Menus are nested and share the expanded source choices. Parallel rates are conditional assumptions. Staircases use only the sampled sizes, with $L = 48$ the sweep endpoint.

5 Direct step checks and finite-patch time dependence

We compute the exact reference evolution in the $N_\uparrow = N_\downarrow = 8$ sector of the 4×4 OBC lattice. Its $\binom{16}{8}^2 = 165{,}636{,}900$ amplitudes fit a matrix-free calculation with four complex state vectors, using about 10 GB of memory. At $\tau = 1$, the central doublon at coordinate $(2, 2)$ is 0.1452313819. The sampling convention remains $D_* = 0.1$ and the absolute targets remain 0.001 and 0.01; we do not silently redefine the requested precision using the newly calculated signal.

The plaquette coefficient $|D_r - D_{\text{exact}}|r^2$ changes from 0.262431 at $r = 16$ to 0.261828 at $r = 32$, a 0.23% difference. It is about 20% above the 3×3 value. The termwise coefficient changes from 0.188386 to 0.187732, a 0.35% difference, and is about 2.45 times its 3×3 value. The 4×3 coefficients, 0.2453 and 0.1392, lie between the square-patch results, as Figure 7 shows. The earlier 10% planning margin therefore does not cover even this $\tau = 1$ transfer. Exact finite-lattice evolution resolves that specific uncertainty; it does not establish an infinite-lattice coefficient.

Validate before reallocating. The initial 4×4 resource sensitivity interpolated between measured steps using C/r^2, with a 10% margin. We now directly evaluate every selected step count for the headline reference designs: effective and two PEC NISQ models, reference STAR and hybrid, and reference, single-fast-block, and four-fast-port FT. The two precision targets give 16 allocations sharing nine circuits. All satisfy

$$|D_r - D_{\text{exact}}| + 10^{-10} \le \epsilon f_T^{\text{saved}}, \tag{11}$$

using their *original* Trotter fraction f_T^{saved}. The measured errors use 89.3%–90.9% of these allowances. No change to statistics or another error component is needed to pass these checks.

We then separately reoptimize with the measured Trotter error and the same 10^{-10} numerical allowance. Table 1 shows the strict estimates at each originally selected r. Reference FT falls from 1.88 to 1.66 million qubits, and the single fast block from 1.16 to 0.99 million, because the smaller charged Trotter error leaves more allowance for sampling. STAR and hybrid remain single-copy

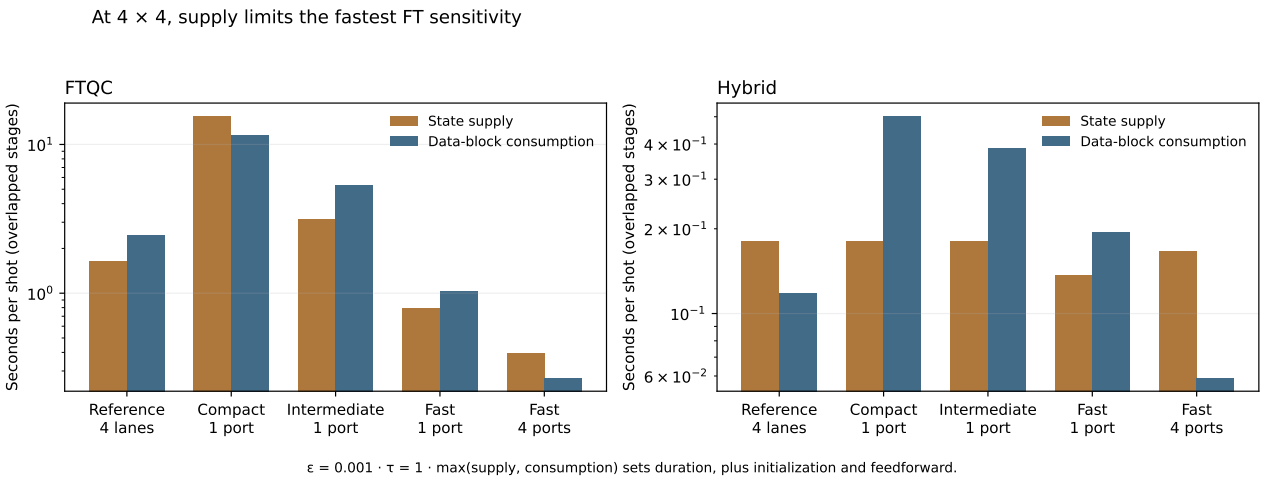


Figure 6: Supply and consumption durations at strict 4×4, $\tau = 1$, with each layout's budgets and source count reoptimized. These stages overlap: their maximum enters shot time. The fastest FT sensitivity becomes supply-limited, while the hybrid also pays for its analog rotations.

Design	Saved r	Saved Q	Measured-error budget Q
NISQ effective	21	72	72
STAR reference	33	71,442	71,442
Hybrid reference	41	73,348	73,348
FT reference	33	1,875,882	1,655,190
FT single fast block	29	1,155,490	988,112
FT four fast ports	33	385,052	385,052

Table 1: Direct 4×4 checks at $\tau = 1$, $\epsilon = 0.001$. Both columns use the same step count, hardware, and source menu. The last column reallocates the smaller measured Trotter error before minimizing resources. Effective NISQ retains its assumed mitigation bias and variance; four fast ports retain the uncompiled throughput assumption.

calculations. The saved Trotter allowances were already adequate; these reductions come from using their remaining allowance. Searching all measured steps gives the same headline areas, sometimes at a different r.

Checking the selected NISQ count $r = 21$ removes the earlier artificial restriction to $r = 8, 16, 32$, which had inflated its measured-menu cost to 2196 qubits. This validates the Trotter component only: plaquette accuracy is still transferred to the experimental gate-count proxy, and mitigation bias remains unmeasured. At loose precision the two-qubit PEC scenario also fits one 36-qubit copy at $r = 8$, whereas its strict estimate exceeds 10^{12} qubits. The full-noise PEC proxy is much more costly. These differences remain visible in the saved tables and should not be read as an architecture-independent NISQ limit.

Longer time and the capacity sensitivity. We extend the same exact 4×4 reference and both product formulas to $\tau = 2, 4$. At each time, the initial counts are $8\tau, 16\tau, 32\tau$; further doubling is required if the two finest coefficients disagree by more than 5%. Figure 8 compares the converged tails with the smaller patches. The observed site remains $(2, 2)$ and each product formula is compared with exact evolution on that same lattice. The converged-pair coefficients on 4×4 are 1.096 and 0.775 for plaquette and termwise evolution at $\tau = 2$, and 4.193 and 3.029 at $\tau = 4$. The finest pairs agree within

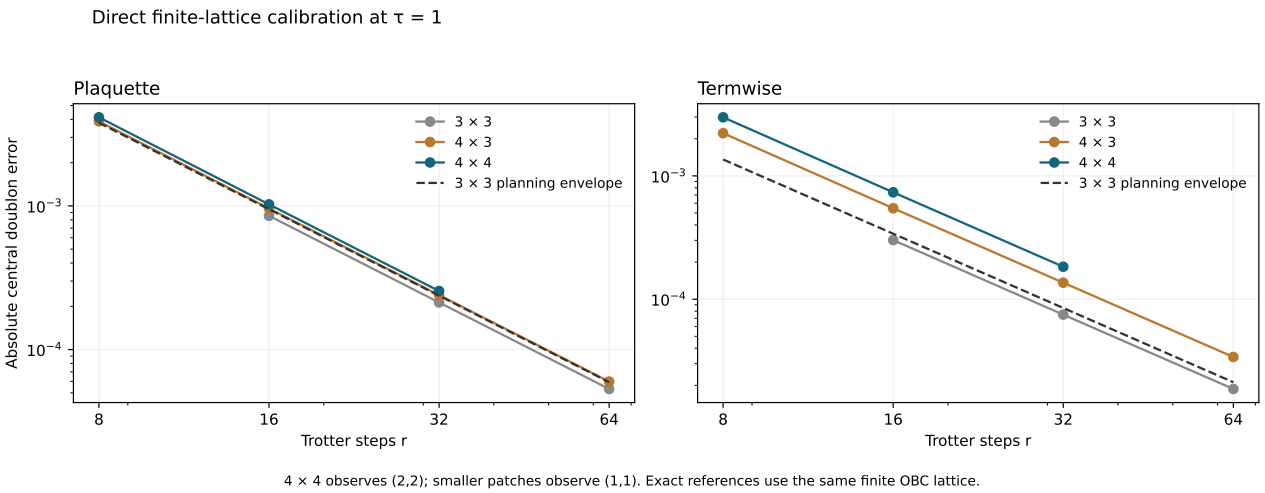


Figure 7: Direct Trotter errors at $\tau = 1$ on three finite patches, with exact references on each same lattice. Both orderings exhibit r^{-2} convergence, with coefficients that depend on the patch. The dashed line is the previous 3×3 planning envelope, including its 10% margin. The 4×4 observable is at $(2, 2)$; smaller patches use $(1, 1)$. This comparison calibrates the finite 4×4 example and does not certify large-L or long-time transfer.

0.36% in all four cases. The $\tau = 4$ plaquette value is close to 4×3 and below 3×3; the termwise value exceeds both smaller patches. A single multiplicative finite-size correction would miss this dependence on ordering and time.

To isolate this physics input, we keep the reference delivery assumptions and the expanded common source menu fixed and reoptimize the resource budgets under two coefficient policies. The first is the earlier 3×3 envelope. The second retains that envelope and adds all measured 4×3 and 4×4 coefficients through the requested time, including finite-step corrections, with the same 10% margin. The $L = 2$ point keeps its own calibration in both cases. Figure 9 reports the resulting capacity sensitivity. It does not replace a finite-patch assumption with a bulk error bound: the new curves still transfer coefficients to $L > 4$, and their optimized step counts have not all been checked directly. No 4×4 coefficient is inferred at $\tau = 8$. The additional patches move several discrete thresholds while retaining the strong precision dependence in this reference model. For example, at $\tau = 2$, $Q = 10^6$, and strict precision, effective NISQ falls from $L = 4$ to $L = 2$, whereas reference STAR and hybrid remain at $L = 4$. This is a sensitivity to the assumed mitigation and finite-patch models; it is not a general NISQ limit.

6 The fixed-device question

The same physical qubits can buy more simultaneous samples, stronger logical protection, or faster delivery of non-Clifford operations. We want to determine which use gives the largest accessible dynamics problem, and which combinations can improve on the three reference approaches. Figure 10 shows the resource roles. The comparison includes physical probabilistic error cancellation (PEC) [8], the original and small-angle STAR mechanisms [1, 5], and surface-code computation with explicit magic-state supply [7, 6]. Campbell's Hubbard construction provides a concrete algorithmic baseline [2]; recent partial-FT resource studies [9] and a performed fermionic-dynamics experiment [10] provide comparison points with distinct hardware assumptions.

We treat these as specified implementations and estimators. A poor physical PEC curve is not

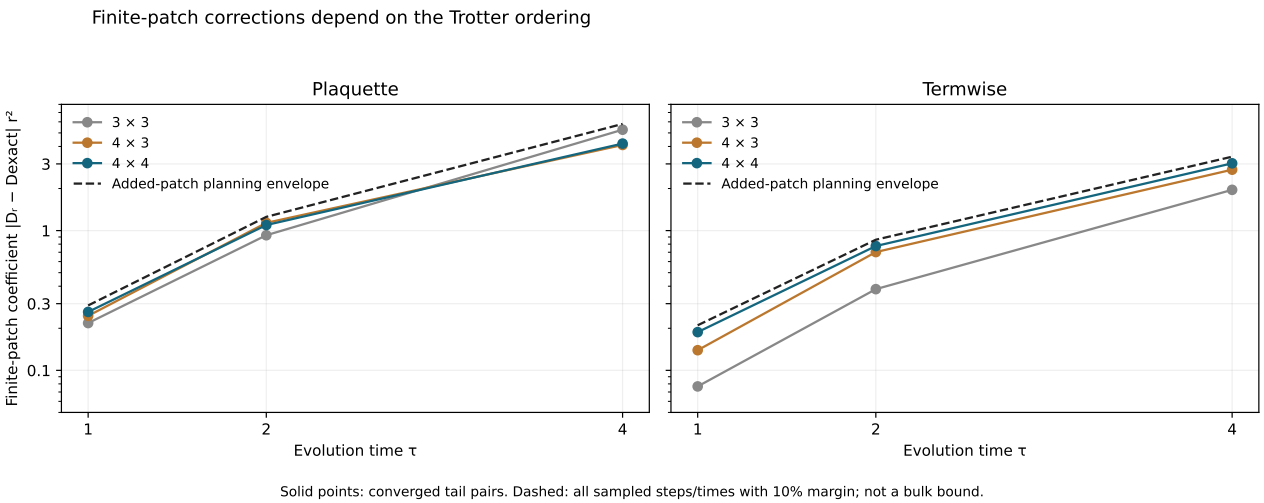


Figure 8: Ordering-specific Trotter coefficients through $\tau = 4$ on three finite patches. Solid points use converged tail pairs; the dashed envelope also covers measured coarser-step coefficients and earlier sampled times, with a 10% margin. The larger patches have only three time nodes; unsampled time peaks and larger-lattice corrections remain possible.

an impossibility theorem for NISQ, and an encoded curve based on reference patch timing is not a validated physical floorplan. The purpose of the analytical laws is to expose the quantities whose measurement or optimization can change the architecture ranking. CUDA-Q Logical supplies explicit circuit and synthesis artifacts [11]; physical encoded timing is kept separate when its implementation or verification is incomplete.

The similarity of the encoded data patches is physical: both approaches protect their Clifford operations. The main distinction is how they supply non-Clifford rotations, including the sequence of operations in time (Figure 13). STAR teleports an angle-specific resource state and mitigates residual rotation errors. Our FTQC reference instead synthesizes the rotation from Clifford gates and high-fidelity T states. In the original STAR preparation protocol, a surgery ancilla can also serve as the preparation patch when distance and physical error rate permit adequate acceptance [5, Sec. II A 2]. Preparation, any patch growth, and retries still occupy space and time. Dedicated injection lanes are therefore a scheduling choice in our cost model, not a defining STAR requirement. Conversely, FTQC supply may be compact or shared in time; neither approach has a universal ratio of data area to preparation area.

7 Task, conventions, and precision

The problem has $N = L^2$ Hubbard sites, $2N$ spin modes, and open boundaries. The physical device has a separate area budget Q. In units where energies are measured in J, we simulate

$$H/J = -\sum_{\langle ij\rangle,\sigma} (c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.}) + u \sum_i (n_{i\uparrow} - \tfrac{1}{2})(n_{i\downarrow} - \tfrac{1}{2}), \qquad u = U/J = 4. \tag{12}$$

The symmetric onsite form differs from $U n_\uparrow n_\downarrow$ only by a known phase in a fixed-number sector. We prepare the half-filled Néel product state and measure the central doublon $D = \langle n_{i\uparrow} n_{i\downarrow} \rangle$ at $\tau = Jt/\hbar$. Doublon avoids the symmetry-pinned charge-density observable of this quench.

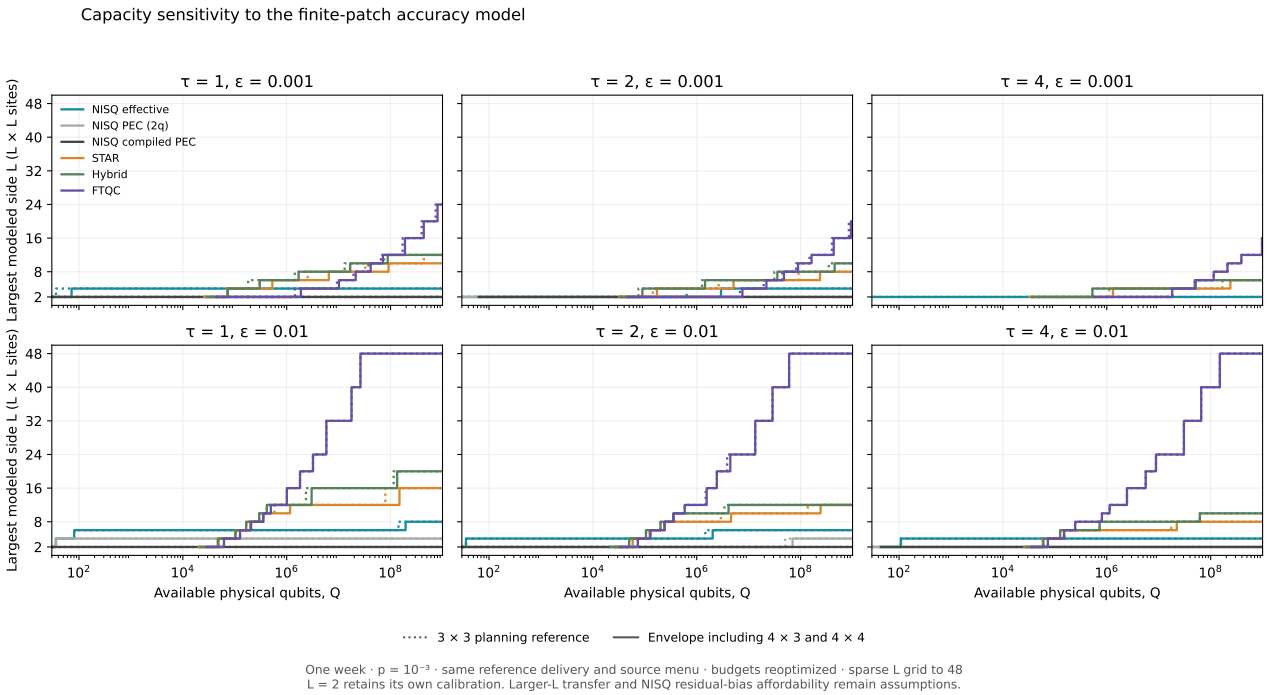


Figure 9: Largest modeled lattice side versus available qubits, changing only the finite-patch accuracy model. Dotted curves use the original 3×3 envelope; solid curves also include the 4×3 and 4×4 measurements. Delivery and source menus are identical, with error allocations reoptimized. Both policies retain direct $L = 2$ calibration. NISQ effective mitigation remains an assumed envelope; two-qubit-only PEC and the full compiled-noise PEC proxy are shown alongside it. All curves use a one-week cutoff, $p = 10^{-3}$, and the same sparse size grid through $L = 48$. A missing curve means that no sampled lattice fits that panel's qubit and time budgets.

Quantity	Reference value or convention
Connectivity	Square array, nearest neighbors
Gate / measurement / reset	10 / 100 / 100 ns
Physical two-qubit / one-qubit error	$p = 10^{-3}$ / $p/10$
Idle channel	Depolarizing error p per 100 ns
Encoded cycle / classical feedforward	240 / 200 ns, reference assumptions
Runtime cutoff	$B = 604800$ s, one week
Main precision	1% relative at provisional $D_* = 0.1$: $\epsilon = 0.001$
Statistical confidence	$1 - \delta = 0.95$
Rotation convention	$U_P(\theta) = e^{-i\theta P}$; CUDA-Q angle $\phi = 2\theta$

The native physical compilation assumes arbitrary-angle R_{ZZ} and one-qubit pulses at the stated gate duration. The noise channels and encoded cycle are explicit model choices, not a characterization of a fabricated device. In particular, the surface-code suppression fit has not been calibrated to the same physical idle and syndrome schedule.

The target is $\epsilon = \eta|D|$ with $\eta = 0.01$. Resource curves use the declared signal D_*, rather than silently treating it as calculated dynamics. Campbell's energy precision proportional to N corresponds to fixed absolute precision in energy density [2]. Our local D is already intensive: fixed absolute error in D does not itself become stricter with size. Relative precision is useful when the signal magnitude is the relevant scale. Strict relative precision at zero signal needs a separate prescription; a production

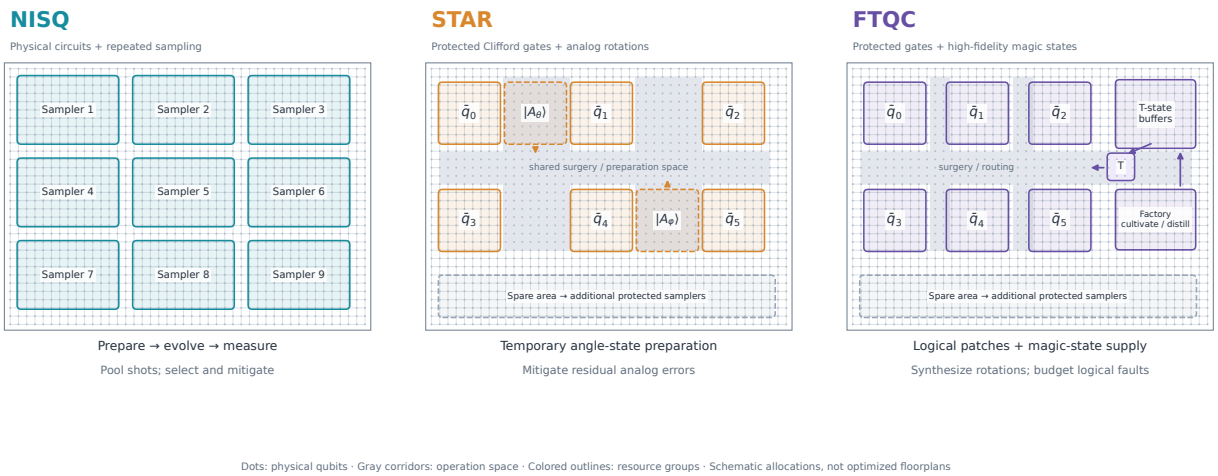


Figure 10: Three allocations of a common sea of physical qubits. NISQ uses many unprotected circuit replicas. STAR and FTQC share protected data patches and lattice surgery. Dashed orange patches show temporary angle-state preparation in STAR surgery ancillas, a possible allocation rather than a dedicated factory requirement [5]. FTQC supplies high-fidelity T states to a synthesized sequence of logical operations. Every dot denotes a physical qubit; each encoded data patch contains many such qubits. The drawings explain resource roles and are not numerical floorplans or equal-capacity implementations of a particular Hubbard lattice.

estimator with unknown nonzero D would require a pilot or sequential confidence procedure. We retain $\epsilon = 0.01$ absolute as a separately labeled sensitivity.

We allocate fractions $(0.5, 0.25, 0.1, 0.075, 0.075)$ of ϵ to statistical, algorithmic, logical, synthesis, and magic-state errors. For a spatial patch Ω and product formula with r steps,

$$|\widehat{D} - D_{\text{target}}| \le \epsilon_{\text{stat}} + |D_{\Omega,r} - D_{\Omega}| + |D_{\Omega} - D_{\text{target}}| + \epsilon_{\text{logical}} + \epsilon_{\text{synth}} + \epsilon_{\text{magic}}. \tag{13}$$

The truncation scenario splits the existing 0.25ϵ algorithmic reserve equally between Trotter and spatial bias. It does not add a new allowance. Mitigation assumes exact channel knowledge; a residual calibration bias would need its own allocation within Eq. (13). No unverified postselection bias reduction is credited in a resource curve.

8 Sampling, replication, and maximum lattice size

For inverse channels $\mathcal{E}_j^{-1} = \sum_a c_{ja}\mathcal{P}_{ja}$, define $\gamma_j = \sum_a |c_{ja}|$, $\Gamma = \prod_j \gamma_j$, and $\chi = \log \Gamma^2$. Centering the doublon observable as $D - I/2$ puts the PEC sample in $[-\Gamma/2, \Gamma/2]$. Hoeffding's bound then gives

$$n \ge \frac{\Gamma^2}{2\epsilon_{\text{stat}}^2} \log \frac{2}{\delta}, \qquad N_0 = \left\lceil \frac{\log(2/\delta)}{2\epsilon_{\text{stat}}^2} \right\rceil. \tag{14}$$

At strict reference precision, $N_0 = 7377759$. This is a sufficient common sampling convention, not a claim that every estimator needs this many shots.

Let q_a and t_a be the physical area and time of one complete sampler, and let a be whole-circuit acceptance. With $m = \lfloor Q/q_a \rfloor$ replicas,

$$T_{\text{answer}} \simeq \frac{N_0 e^{\chi} t_a}{am}, \qquad Q_{\min}(L) = q_a \max\left(1, \left\lceil \frac{N_0 e^{\chi} t_a}{aB} \right\rceil\right). \tag{15}$$

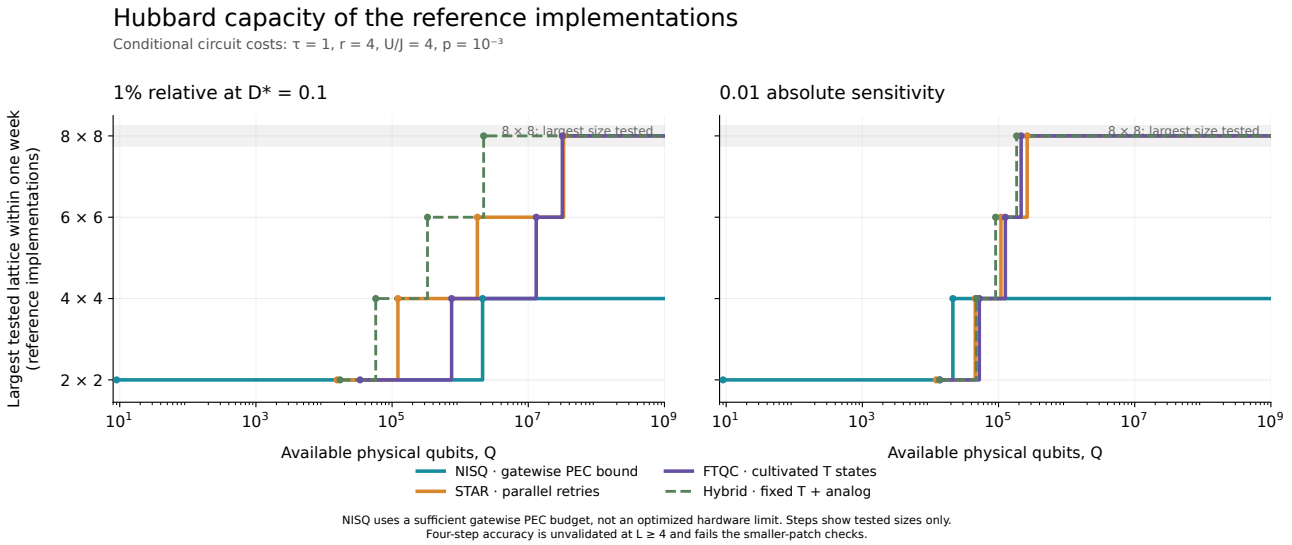


Figure 11: Largest tested Hubbard lattice within one week versus available physical qubits. Left: the accepted relative-precision reference. Right: the loose absolute-precision sensitivity. NISQ uses a sufficient gatewise-PEC sampling bound on the compiled observable cone at nominal idle noise; it does not represent optimized NISQ capacity. STAR chooses among one, two, four, and eight reference injection lanes; FTQC and hybrid use the packed cultivation supply scenario with zero additional handoff error. These are conditional $r = 4$ costs, not accuracy-qualified dynamics capacities. That step count fails the small square-patch checks, and $L \geq 4$ accuracy is uncalibrated. The NISQ compiler includes exact observable reductions that the encoded workloads have not yet received. Only $L = 2, 4, 6, 8$ is tested; the upper band marks that limit.

The code excludes a sampler whose single attempt exceeds B. Resource-state rejection is charged in delivery time, not counted again as whole-circuit rejection. Integer replicas are retained where they matter. Packing is an area-budget reference; a global square placement, calibration overhead, and control bandwidth are not yet modeled in the encoded curves. For PEC, $Q_{\min}$ is the minimum replica area that meets this sufficient sampling bound for the chosen implementation. It is not a lower bound on the qubits needed by another estimator, or even a tight cost for this one.

The principal output is now

$$L_{\max}^{\text{tested}}(Q) = \max\{L \in \mathcal{L}_{\text{tested}} : \min_{s} Q_{\min}(L, s) \leq Q\}, \tag{16}$$

where s labels allowed distances, supply configurations, and lane counts. This is a discrete upper envelope, not a smooth inverse fit. No feasible tested lattice is left blank, and reaching the largest tested size is marked as a survey boundary. We do not interpolate untested sizes or call that boundary a physical saturation. The companion output $\tau_{\max}(Q, L)$ follows by sweeping evolution time under the same contract; it has not yet been accuracy-qualified.

For a fixed circuit and estimator, Eq. (15) gives the exposure limit

$$\chi \lesssim \log \frac{amB}{N_0 t_a}. \tag{17}$$

Increasing hardware or runtime therefore buys logarithmic exposure, even though it can move a discrete lattice-size threshold appreciably. This is the mitigation wall of the specified strategy, not a universal NISQ law.

9 The common algorithm and its angle inventory

9.1 Termwise second order

In a fixed-number sector, the single-Z terms of the onsite interaction are a known phase. With $E = 2L(L-1)$ spatial bonds, the remaining Jordan–Wigner Hamiltonian has

$$K = 4E + N = 9N - 8L, \qquad \lambda = \sum_{\ell} |h_\ell| = 2E + \frac{|u|N}{4} = 5N - 4L. \tag{18}$$

There are four hopping Pauli strings per spatial bond, each with coefficient magnitude $1/2$, and one onsite ZZ per site, with magnitude $|u|/4$. Our convention is $U_P(\theta) = e^{-i\theta P}$; thus $R_z(\phi)$ has $\phi = 2\theta$, and a T gate has $|\theta| = \pi/8$ up to a global phase.

For a symmetric product over the K terms, choose onsite terms at both ends of the ordering and merge adjacent identical rotations at the center and between steps. Before further commuting or Clifford simplifications,

$$R_{\text{term}} = r(2K-2) + 1, \qquad \Theta_{\text{term}} = \tau\lambda. \tag{19}$$

The count follows from $r(2K-1)$ rotations after center merging and $r-1$ additional boundary merges. Every coefficient has total positive duration τ, proving the angle identity. Signed higher-order stages instead multiply the angle sum by their total absolute stage weight. The histogram, not merely Θ, is retained for channel evaluation.

9.2 Campbell plaquette second order with open boundaries

Campbell partitions hopping into disjoint pink and gold tiles and applies

$$S_2(\Delta) = e^{-i\Delta H_I/2} e^{-i\Delta H_p/2} e^{-i\Delta H_g} e^{-i\Delta H_p/2} e^{-i\Delta H_I/2}, \qquad \Delta = \tau/r. \tag{20}$$

We adapt the periodic construction to open boundaries. The pink partition has $P = L^2/4$ complete plaquettes. The shifted gold partition has $G = (L/2-1)^2$ complete plaquettes and $B = 2(L-2)$ isolated boundary bonds. Each bond belongs to exactly one partition, since $4(P+G) + B = E$.

A single-spin four-site hopping cycle has eigenvalues $\{2, 0, 0, -2\}$. Campbell's circuit implements its evolution with two variable-angle Pauli rotations and eight fixed T gates [2, Appendix E.1]. An isolated bond needs two commuting hopping Pauli rotations per spin. For $L \geq 4$, including both spin species and exact onsite boundary merging, this gives

$$R_{\text{plaq}} = N(r+1) + 4r(2P + G + B), \tag{21}$$

$$T_{\text{fixed}} = 16r(2P + G), \tag{22}$$

$$\Theta_{\text{dyn}} = \tau \left[\frac{|u|N}{4} + 4(P+G) + 2B \right]. \tag{23}$$

For $L = 2$ the gold partition is empty: the two pink half steps merge, so $2P$ becomes P in the count formulas. The angle formula is unchanged. At $L = 4$, $\tau = 1$, $r = 4$, these formulas give $(R_{\text{plaq}}, T_{\text{fixed}}, \Theta_{\text{dyn}}) = (288, 576, 44)$.

The compiled total angle includes $T_{\text{fixed}}\pi/8$. Consequently, the invariance of $\tau\lambda$ under refinement in Eq. (19) does not apply to all non-Clifford operations in Eq. (20). The fixed basis changes are an algorithm-dependent STAR exposure even when the variable rotations have no injection floor. Distilling those fixed angles is an immediate hybrid candidate. Hamming-weight phasing, cancellations between neighboring basis changes, and angle extraction remain further compiler optimizations. Campbell neglects fermionic-swap Clifford costs in his non-Clifford metric; our native nearest-neighbor compilation explicitly restores this motion.

10 Physical computation: gates, locality, and idle exposure

For a k-qubit depolarizing location with total nonidentity-Pauli probability p_k, set $s_k = 1 - 4^k p_k/(4^k - 1)$. The inverse has identity coefficient $1 + p_k/s_k$ and $4^k - 1$ coefficients $-p_k/[(4^k - 1)s_k]$, giving

$$\gamma_k = 1 + 2p_k/s_k, \qquad \chi_{\text{phys}} = \sum_{\ell \in \text{cone}} 2 \log \gamma_{k_\ell} \simeq 4pG_2 + 4(p/10)G_1 + 4p_{\text{idle}} G_{\text{idle}} + \chi_{\text{SPAM}}. \tag{24}$$

Preparation and readout bit flips use $\gamma = 1/(1 - 2p)$. The continuous-time idle convention composes as

$$p_{\text{idle}}(\Delta t) = \frac{3}{4}\left[1 - (1 - 4p/3)^{\Delta t/(100\,\text{ns})}\right]. \tag{25}$$

Only channels in the compiled measurement's backward dependency cone enter the local PEC product. CPTP channels outside that cone cannot affect its expectation. This sufficient cone can still exceed the observable support needed at a specified tolerance.

10.1 Fermionic blocks and parallel motion

Two-mode hopping, density evolution, and fermionic Hadamards are extracted from the plaquette workload while preserving its noncommuting stage order. A number-conserving Gaussian gate has odd-sector matrix $G \in U(2)$, vacuum amplitude one, and double-occupation amplitude $\det G$. An Euler decomposition of G therefore compiles to local Z rotations and at most two native R_{ZZ} gates. In particular,

$$V(\theta) = e^{-i\theta(XX+YY)}, \qquad F_{\text{swap}} = (S \otimes S)V(\pi/4), \tag{26}$$

so a swap followed by hopping can fuse into the same pair of entanglers with a changed angle. Products of Gaussian gates on the same adjacent pair fuse through operations on disjoint wires. A commuting density rotation adds at most one entangler.

The compiler gathers independent plaquette groups into intervals of a live fermionic order using parallel odd-even transposition sorting. All swaps are adjacent physical fermionic swaps on a snake embedded in the square grid [12]. It then executes disjoint groups concurrently and tries six initial-order/local-route choices. A preparation permutation is free for the specified occupation product state, while the final wire map and Fock signs remain explicit. Using only snake edges leaves room for a better two-dimensional compiler. The trapped-ion experiment's all-to-all connectivity and free ordinary-SWAP relabeling are a different compilation assumption [10].

The first onsite stage is a scalar on the initial Fock state, and the last onsite stage commutes with the measured doublon. Removing them is exact for this task. A reverse traversal through even fermionic blocks then removes blocks on disjoint support. These reductions are available to every arm; the current encoded resource curves retain the original full logical workload.

$L = 4,\ \tau = 1,\ r = 4$	Entanglers	10 ns layers	Local χ
Greedy fused route with cone	3149	2187	35.297
Parallel route, full evolution	3204	1074	22.214
Parallel route with observable reductions	2648	964	20.017

The final row has $t_a = 9.84\ \mu$s and exposure

$$\chi = 10.5927_{2q} + 4.1201_{1q} + 5.1681_{\text{idle}} + 0.1361_{\text{SPAM}}. \tag{27}$$

Its strict-precision threshold is $Q_{\min} \simeq 2.13 \times 10^6$, versus 12168 with zero idle on the same circuit. Idle has decreased substantially but remains material. The gain combines scheduling, placement, fusion, and observable algebra; zero idle is a sensitivity rather than a demonstrated alternate schedule.

10.2 What the million-qubit estimate means

The 4×4 circuit occupies 32 active qubits, packed into a 36-qubit square. At the strict precision, the reference assigns $\Gamma^2 = e^{20.017} \simeq 4.94 \times 10^8$ and a sufficient budget of $N_0\Gamma^2 \simeq 3.64 \times 10^{15}$ shots. At 9.84 μs per attempt, this requires 59239 copies to fit within one week, giving $Q_{\min} = 2132604$. The large area is entirely a sampling-throughput cost. For the same circuit, duration, and precision, removing idle noise gives $Q_{\min} = 12168$; retaining only two-qubit noise gives $Q_{\min} = 180$. These are noise ablations, not demonstrated improvements or alternative predictions for the declared device.

Experiments require an additional calibration of the NISQ model. Phasecraft and Quantinuum simulated a 7×4 lattice with 56 qubits, using training with fermionic linear optics (TFLO) and time-series smoothing [10]. The Willow study reached 6×6 with 72 fermionic-mode qubits, using TFLO, symmetry selection, and other application-specific processing [4]. Our gatewise PEC bound does not describe these estimators. Nor do those demonstrations establish our particular 0.001 absolute-error contract at every interacting data point. Their compilation is also a relevant benchmark: Willow's published 4×4 formula gives 1620 CZ gates at four steps, where our reference uses 2648 native R_{ZZ} gates. This comparison needs the respective gate implementations and schedules, not only a gate-count ratio. We therefore retain the current curve as a specified PEC reference; the resource frontier for experimentally informed physical estimators remains unestablished.

10.3 Exact circuit cones and physical truncation

For the original plaquette block ordering, the sufficient interior cones checked at $r = 1, 2, 4$ contain $(4r + 2)^2 = 36, 100, 324$ sites. The $r = 1$ circuits at $L = 8, 12, 16, 20$ all compile to 2001 native entanglers on 72 modes in that implementation. Ordered termwise sweeps can instead have global strict dependencies even at one step. Neither observation establishes the smallest Hamiltonian region needed at finite tolerance.

At fixed evolution time a local-observable truncation may eventually remove size dependence, whereas a time family such as $\tau \propto L$ probes growing physical reach. A truncation must receive a bias allowance in Eq. (13); it cannot be inferred solely from the strict circuit cone. Section 14 begins the finite-patch check and states what its outer boundary leaves unresolved.

11 Encoded protection and the logical-fault budget

Both encoded arms use the declared per-patch, per-cycle suppression reference

$$p_L(d) = A(p/p_{\text{th}})^{(d+1)/2}, \qquad A = 0.1, \quad p_{\text{th}} = 0.01. \tag{28}$$

The alternative $(A, p_{\text{th}}) = (0.07, 0.0039)$ is motivated by the original STAR fit [1]. Its time convention and circuit noise must be reconciled before it becomes a calibration of this device. We use 240 ns per syndrome cycle and roughly $2d^2$ physical qubits per tile as reference accounting. Figure 12 distinguishes physical qubits, logical patches, and operation/supply space.

For V_L patch-cycles per attempt, $q_L \le V_L p_L(d)$ by a union bound. A $[0, 1]$ unmitigated observable has fault bias at most q_L. A uniform branch-wise bound for centered PEC gives the sufficient worst-case Γq_L. Thus the reference chooses the smallest admissible odd d with

$$V_L p_L(d) \le \epsilon_L/\Gamma, \qquad d \gtrsim \frac{2\log(AV_L\Gamma/\epsilon_L)}{\log(p_{\text{th}}/p)} - 1. \tag{29}$$

The implementation solves the discrete self-consistency problem because duration and volume also depend on d. Faults are budgeted per attempt; the full sampling campaign is not required to contain zero logical faults. Commuting, branch-independent fault channels can justify removing Γ locally.

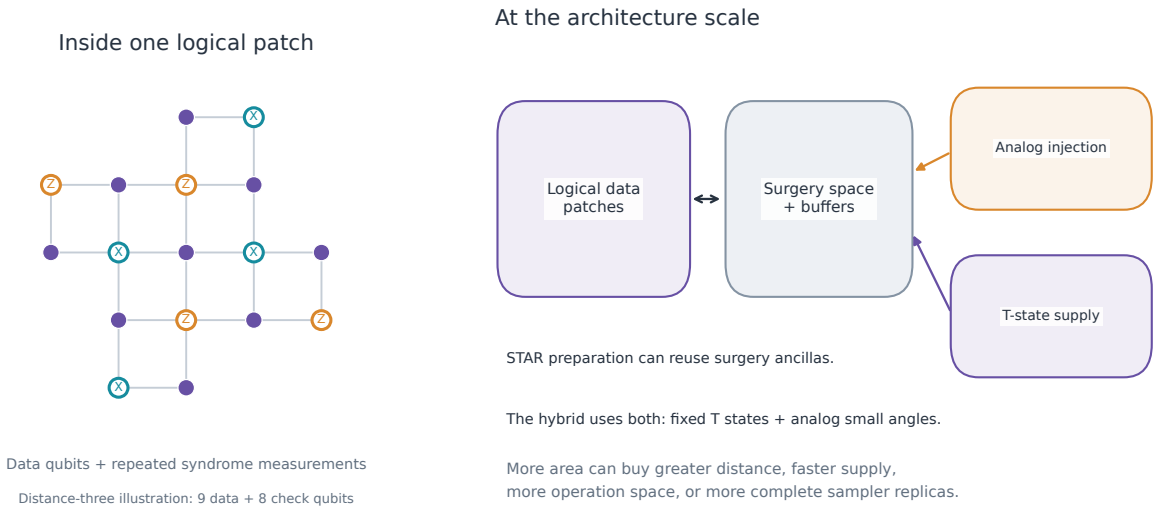


Figure 12: Physical and logical resource scales. Left: a distance-three rotated-code illustration with nine data and eight check qubits; links are nearest-neighbor data–check interactions. Right: logical data interacts through operation space with analog injection or T-state supply. STAR and FTQC use different supply paths; the hybrid uses both. The diagram shows functional roles: STAR preparation can occupy surgery ancillas rather than a separate supply region. This illustration is not a choice of distance or a calibrated surgery schedule for the resource curves.

Section 12.1 checks this condition for a declared logical injection instrument, and Appendix B explains why independence alone is insufficient.

12 STAR: angle exposure, floors, and delivery

For independent residual dephasing q_j after completed rotations,

$$\Gamma^2 = \prod_j (1-2q_j)^{-2}, \qquad \chi_{\mathrm{STAR}} = -2\sum_j \log(1-2q_j) \simeq 4\sum_j q_j. \tag{30}$$

Original STAR uses the post-RUS reference $q_0 = 4p/15$ [1]. The small-angle reference uses $q_j = \alpha p|\theta_j| + f$, with $\alpha = 1.5$ in our angle convention and a declared floor $f = 10^{-5}$ [5, 9]. The floor is a sensitivity motivated by injection and growth, not a measured error floor at this operating point. Fixed $\pi/8$ rotations retain q_0; we do not extrapolate the small-angle fit to them. Consequently,

$$\chi_{\mathrm{STAR}} \simeq 4\alpha p\Theta_{\mathrm{dyn}} + 4fR + \frac{16p}{15}T_{\mathrm{fixed}}. \tag{31}$$

The implementation evaluates the logarithmic product in Eq. (30); the physical channel models themselves remain approximations. Their three exposures have different responses to compilation and refinement. In particular, $fR \sim \alpha p\Theta_{\mathrm{dyn}}$ marks a floor/angle competition, while repeated basis changes can dominate separately.

The serial reference allocates $n_{\mathrm{data}} = \lceil 1.5(2N+6) \rceil$ tiles and six injection tiles. It charges 18 code clocks per rotation, with one clock equal to d syndrome cycles, plus startup, readout, and feedforward. This is a delivery model. A small-angle error law does not itself establish its acceptance, throughput, or footprint. The explicit retry refinement below adds both parallelism and its area cost.

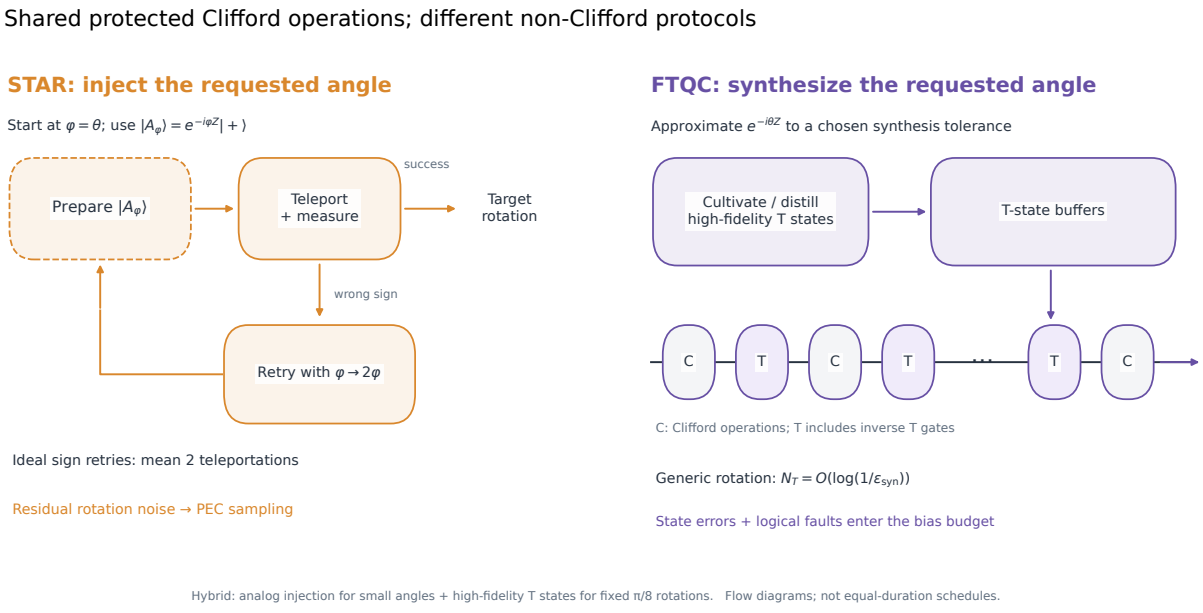


Figure 13: Two implementations of the same target rotation, in the convention $e^{-i\theta Z}$. STAR prepares an angle-specific resource state and uses teleportation with angle-doubling corrections [1, 5]. Ideal sign outcomes require two teleportations on average; this does not include rejected preparation attempts or their cost. FTQC consumes high-fidelity T states in a Clifford+T approximation, whose generic T count grows logarithmically with inverse synthesis tolerance [13]. The hybrid chooses between these paths by operation class. These are protocol flows, not schedules with equal durations or calibrated factory footprints.

12.1 Explicit retries, covariance, and parallelism

The logical injection instrument prepares $|A_\theta\rangle = e^{-i\theta Z}|+\rangle$, applies $\mathrm{CNOT}_{\mathrm{data}\to\mathrm{ancilla}}$, and measures the ancilla in Z. Its data Kraus operators are

$$K_0 = 2^{-1/2} e^{-i\theta Z}, \qquad K_1 = 2^{-1/2} e^{+i\theta Z}. \tag{32}$$

After outcome one, an injection of angle 2θ corrects a successful next branch: $K_0(2\theta)K_1(\theta) = K_0(\theta)/\sqrt{2}$. Repeating this gives the standard angle-doubling protocol underlying the STAR reference [1, 5]. CUDA-Q P0 now contains explicit preparation, CNOT, measurement, conditional retries, and a returned exhaustion flag for caps $1, 2, 4, 8, 24$. This compilation does not supply a surface-code injection or patch-growth implementation.

For a Pauli fault on either wire immediately before or after CNOT, each conditional K remains diagonal or antidiagonal with equal-magnitude entries. Hence

$$K^\dagger K = I/2, \qquad KZ = \pm ZK. \tag{33}$$

The first identity makes the outcome probability independent of the data state; the second makes the conditional map commute with the channel $\mathcal{Z}(\rho) = Z\rho Z$. It therefore commutes with a Z-dephasing inverse, which is a linear combination of $\mathcal{I}$ and $\mathcal{Z}$. Mixtures of such faults and compositions over retries retain this local covariance under the stated branch-independent Pauli model. All sixteen two-qubit Paulis at both locations are checked independently.

This verifies the needed local commutation condition for that declared instrument. It does not establish branch-independent residual noise for physical state preparation, patch growth, measurement-dependent selection, or a mismatched channel inversion. The worst-case Γq_L bias bound therefore

remains the default; q_L is a separately labeled covariance-based reference scenario. The physical STAR schedule still has to justify using it.

For b simultaneous ideal injections with geometric success probability $1/2$ and cap K, the mean batch retry time in units of one attempt is

$$\mathbb{E}[M_b] = \sum_{k=0}^{K-1} \Pr(M_b > k) = \sum_{k=0}^{K-1} \left[1 - (1 - 2^{-k})^b\right]. \tag{34}$$

It approaches 2 for one lane, but is approximately 3.50 for four lanes. Simply dividing the serial duration by four misses the slowest retry. The implemented policy has a barrier between batches, reserves six injection tiles per lane, and assumes nine clocks per attempt, recovering the previous 18-clock serial mean. It forms disjoint batches using full JW supports, not merely disjoint fermionic endpoints. An asynchronous policy could improve this conservative batch schedule.

With R rotations the ideal no-exhaustion probability is $(1 - 2^{-K})^R$. We choose K from a small allocated tail allowance and charge the acceptance penalty. At $L = 4$ and strict precision, two lanes minimize area at $Q = 121716$, versus 166698 for the earlier serial reference. Four and eight lanes reduce attempt time further but lose on area. At loose precision one lane still wins. These results constrain the particular logical mapping and resource model, not the best possible STAR architecture.

13 Full fault tolerance and the hybrid

For R synthesized rotations each with operator error ζ, telescoping unitaries bound the observable bias by $2R\zeta$. We set $\zeta = \epsilon_{\text{syn}}/(2R)$. The deterministic synthesis fit

$$T_R = \lceil 3.02 \log_2(1/\zeta) + 0.77 \rceil, \qquad N_T = T_{\text{fixed}} + RT_R, \tag{35}$$

has the expected logarithmic scaling [13]. At the compiled points, actual CUDA-Q GridSynth words replace this estimate. At $L = 4, \tau = 1, r = 4$, termwise Strang has 889 variable rotations, no fixed T gates, and 61178 synthesized T gates. Plaquettes have 288 variable rotations and 576 fixed T gates, for 19584 synthesized T gates: 66 per variable rotation. Independent matrix reconstruction checks the precision of every distinct emitted word. Whole-program synthesis at $L = 2, 4$ agrees with the class-weighted counts; $L = 6, 8$ use those counts before inter-gate optimization.

13.1 Conventional factory reference

Injected-state error p_{in} is separate from gate error p. The current conventional reference uses $46p/15$, with alternatives retained as sensitivities; this coefficient is not calibrated to our asymmetric $p_{1q} = p/10$ noise model. For a 15-to-1 stage [7],

$$a_1 = 1 - 15p_{\text{in}}, \quad q_1 = 35p_{\text{in}}^3, \quad q_2 = 35q_1^3. \tag{36}$$

One stage reserves twelve tiles, including its output, and takes $11/a_1$ mean code clocks. The deliberately serial two-stage reference reserves two eleven-tile blocks and fifteen storage tiles, so

$$n_F = 37, \qquad c_F = \frac{15(11/a_1) + 11}{1 - 15q_1}. \tag{37}$$

Both stages use the selected code distance, and protected factory faults enter V_L. Correlations, transport, and rejection-tail latency are not resolved by this leading-order mean-service model.

For F factory copies, the FT timing reference is

$$C_{\mathrm{FT}} = 2 + \max(9N_T + 2R, \lceil N_T/F \rceil c_F), \tag{38}$$

$$t_a = C_{\mathrm{FT}} dt_c + N_T t_{\mathrm{ff}} + t_{\mathrm{reset}} + t_{\mathrm{meas}}, \tag{39}$$

$$q_a = 2d^2(n_{\mathrm{data}} + Fn_F). \tag{40}$$

Nine clocks per T is the compact-block consumption reference [7]; the $2R$ term is an extra Clifford-work placeholder. We optimize F over $\{1, 2, 4, 8, 16\}$ together with level and distance, subject to $N_T q_{1,2} \leq \epsilon_{\mathrm{magic}}$. Supply and consumption must be varied together: speeding consumption alone does not help a supply-limited circuit.

13.2 Why the hybrid is a separate candidate

The hybrid supplies the fixed T rotations as high-fidelity states while retaining analog small rotations. It removes the last term of Eq. (31), but pays factory area, supply time, and a magic-state budget amplified by the retained analog PEC reference. At fixed ideal shot budget, a proposed additional protection step improves the wall-clock cost when

$$\Delta \log t_a + \Delta\chi - \Delta \log a - \Delta \log m < 0. \tag{41}$$

This provides a marginal criterion for operation-dependent protection, observable-aware allocation, and mixed rotation delivery. It must include the lost replicas when protection consumes area. The earlier hybrid gain at $L = 4$ reversed under conventional two-level injection supply; that was an amortization effect of the serial reference, not a general disadvantage of the hybrid. Cultivation changes the comparison again, for both FT and hybrid, under the supply assumptions that follow.

13.3 Cultivation for both FT and hybrid supply

Gidney, Shutty, and Jones report cultivated-state error approximately 2×10^{-9} at $p = 10^{-3}$ and, in their Fig. 15, expected volume 6.6×10^4 qubit-rounds for the distance-five protocol with early rejection and sharing [6]. The output is in a grafted matchable code after escape to nominal distance 15; its behavior is not identical to an ordinary distance-15 surface-code patch. Holding, transfer, growth, and consumption require additional accounting.

Our declared packed-supply scenario rounds the expected volume to $V = 10^5$ qubit-cycles, takes footprint $q_f = 1024$, successful latency $\ell_f = 50$ cycles, and charges a $2d$-cycle handoff. These latter choices are supply assumptions, not a compiled factory. The comparison also uses an unshared fixed-slot volume $V = 1024 \times 50/0.01 = 5.12 \times 10^6$. Mapping the paper's uniform physical noise and round definitions to this device's $p_{1q} = p/10$ and 240 ns cycle remains conditional.

For handoff error q_h, define

$$q_{\mathrm{ready}} = 2 \times 10^{-9} + q_h, \qquad s_{\mathrm{raw}} = \max(V/q_f, \ell_f) + 2d. \tag{42}$$

We sweep $q_h = 0, 10^{-9}, 10^{-8}$. Direct consumption is permitted only when

$$N_T q_{\mathrm{ready}} \Gamma \leq \epsilon_{\mathrm{magic}}, \qquad \epsilon_{\mathrm{magic}} = 0.075\epsilon, \tag{43}$$

with $\Gamma = 1$ for FT and the retained conservative analog-mitigation amplification for hybrid. Otherwise one additional 15-to-1 stage is available, with the same reference coefficients used previously [7]:

$$q_{\mathrm{out}} = 35 q_{\mathrm{ready}}^3, \qquad s_{\mathrm{out}} = \frac{15 s_{\mathrm{raw}} + 11d}{1 - 15 q_{\mathrm{ready}}}. \tag{44}$$

The area includes the source footprint, one handoff buffer per source, and twelve further tiles if this stage is used. Surface-memory faults of the data, buffers, and final distillation stage enter the logical budget; the source preparation error already appears in Eq. (43). Supply and consumption run concurrently, so duration contains their maximum, and source copies and code distance are optimized under the week cutoff.

At strict precision and zero additional handoff error, direct unamplified consumption supports at most 37500 states. The compiled $L = 4$ FT count, 19584, fits; the $L = 6$ circuit requires the extra stage. Taking $q_h = 10^{-8}$ forces that stage already at $L = 4$. Cultivation therefore does not remove a distillation level independently of the task budget. In particular, the roughly 10^5 cutoff discussed around TopQAD Fig. 16 counts small-angle rotations at about 100 T states per rotation, not 10^5 T states [9].

$L = 4$, $r = 4$ reference	Q, $\epsilon = 10^{-3}$	Q, $\epsilon = 10^{-2}$
Serial STAR	166698	45486
FT, conventional serial factory	13791250	216890
FT, packed cultivation, $q_h = 0$	748656	52180
Hybrid, conventional serial factory	669714	54150
Hybrid, packed cultivation, $q_h = 0$	57472	47232

The conventional reference uses injected-state error $46p/15$. All rows remain conditional on $r = 4$. The loose-precision FT rows reuse the stricter synthesized words; resynthesis could improve them further. Cultivation changes the supply comparison appreciably, but neither these numbers nor their ranking should be promoted into a result for accurate dynamics.

14 Accuracy, truncation, and noisy observables

The calibration constructs H directly in fixed $(N_\uparrow, N_\downarrow)$ sectors. Sparse matrix exponentiation evaluates $e^{-i\tau H/J}$ without using the compiled gate decomposition. A separate evolution applies

$$U_r(\tau) = \left[e^{-i\Delta H_U/2} e^{-i\Delta H_{\text{pink}}/2} e^{-i\Delta H_{\text{gold}}} e^{-i\Delta H_{\text{pink}}/2} e^{-i\Delta H_U/2}\right]^r, \qquad \Delta = \tau/r, \tag{45}$$

where energies are in units of J. The largest patch is 3×3, with sector dimension $\binom{9}{4}\binom{9}{5} = 15876$. An independent eight-qubit dense Hamiltonian verifies the exact-sector observable; the original compiled plaquette circuit independently verifies the product formula on 2×2. The saved evidence contains 112 Trotter comparisons and 164 exact-curve points over four patches.

Patch, $\tau = 1$	Exact D	Error $r = 4$	Error $r = 16$	Error $r = 64$
2×2	0.1179790	0.0136397	0.00087075	0.00005447
3×3	0.1050849	0.0143939	0.00085350	0.00005317

For these square patches, $r = 64$ is the first tested power of two passing the strict split budget; $r = 16$ passes the loose split budget. These are not minimal-step claims. The asymptotic local error is consistent with r^{-2}, while coarse steps can be nonmonotonic. We also record a budget based on 1% of each actual patch signal, rather than silently replacing the provisional D_* in every resource curve.

The compiled $L = 2$ circuits at $r = 4, 16, 64$ contain respectively $164, 740, 3044$ native entanglers. The selected PEC estimator needs only one nine-qubit square copy at $r = 16$, but approximately 4.73×10^6 qubits at $r = 64$ to meet the week cutoff. Thus a common Trotter refinement can affect

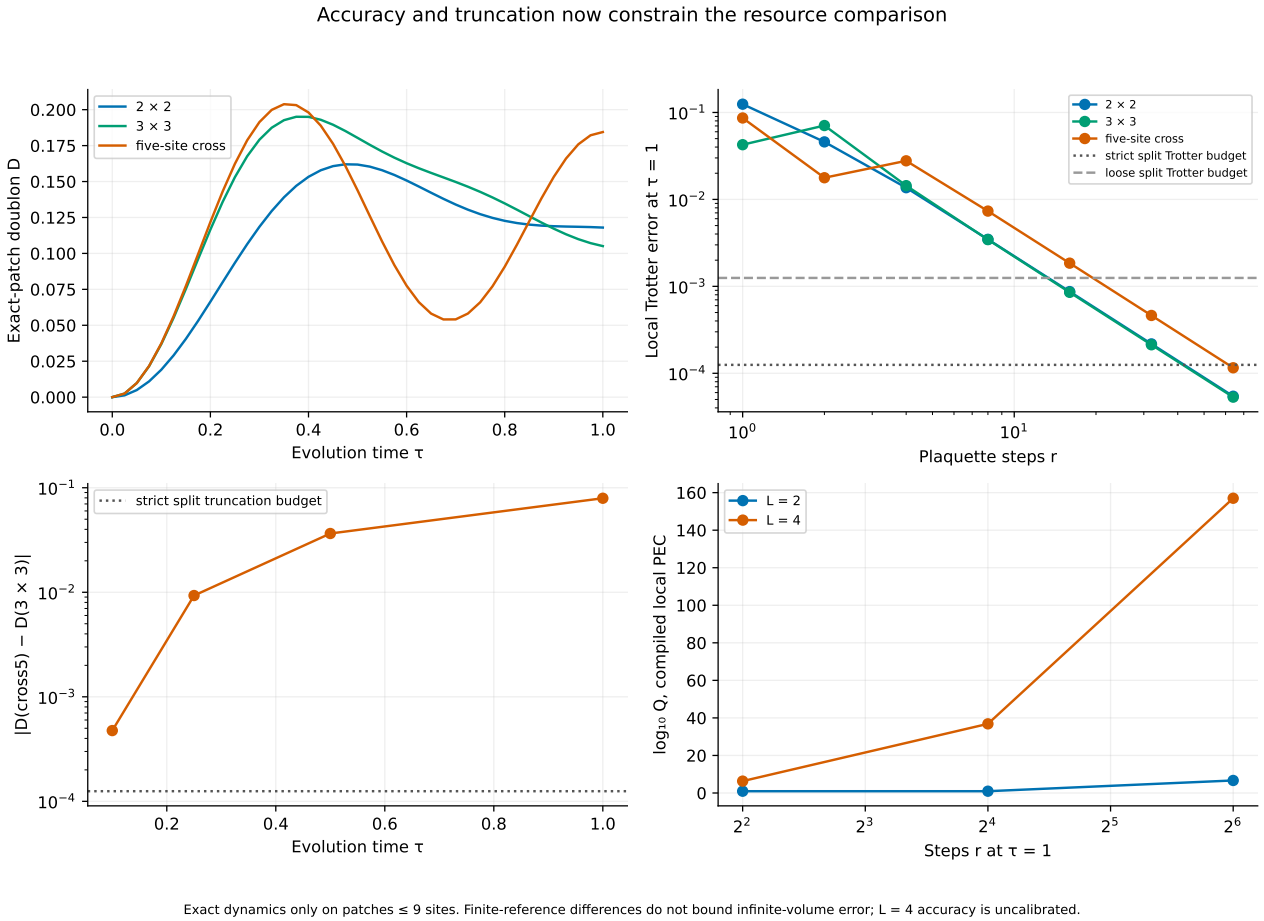


Figure 14: Exact patch dynamics, local product-formula errors, a nested spatial comparison, and the cost sensitivity to r. Each exact curve belongs to its own finite patch. The spatial difference is not an infinite-volume error certificate. The $L = 4$ cost sweep does not assume the small-patch error law holds at $L = 4$.

architecture costs very differently: it increases a physical noise exposure in an exponential, even when its algorithmic error reduction is shared. The corresponding $L = 4$ compilations demonstrate cost sensitivity only; their Trotter error has not been calibrated.

To start testing physical truncation, compare a five-site cross with its enclosing 3×3 square, keeping the same Néel pattern and central observable. Their exact differences are 4.75×10^{-4} at $\tau = 0.1$ and 7.93×10^{-2} at $\tau = 1$. Both exceed the strict truncation allowance. However,

$$|D_{\text{cross}} - D_{\text{target}}| \le |D_{\text{cross}} - D_{3\times3}| + |D_{3\times3} - D_{\text{target}}|, \tag{46}$$

and the second term is unknown. These finite-reference differences neither certify a bulk light cone nor bound the error of the larger patch. Larger patches, controlled local bounds, or both are still needed.

14.1 Postselection: measuring bias as well as acceptance

We propagate the full eight-qubit density matrix through the saved $L = 2$ native circuits. After each k-qubit primitive, the channel is

$$\mathcal{D}_{k,p}(\rho) = (1-p)\rho + \frac{p}{4^k - 1}\sum_{P\neq I} P\rho P. \tag{47}$$

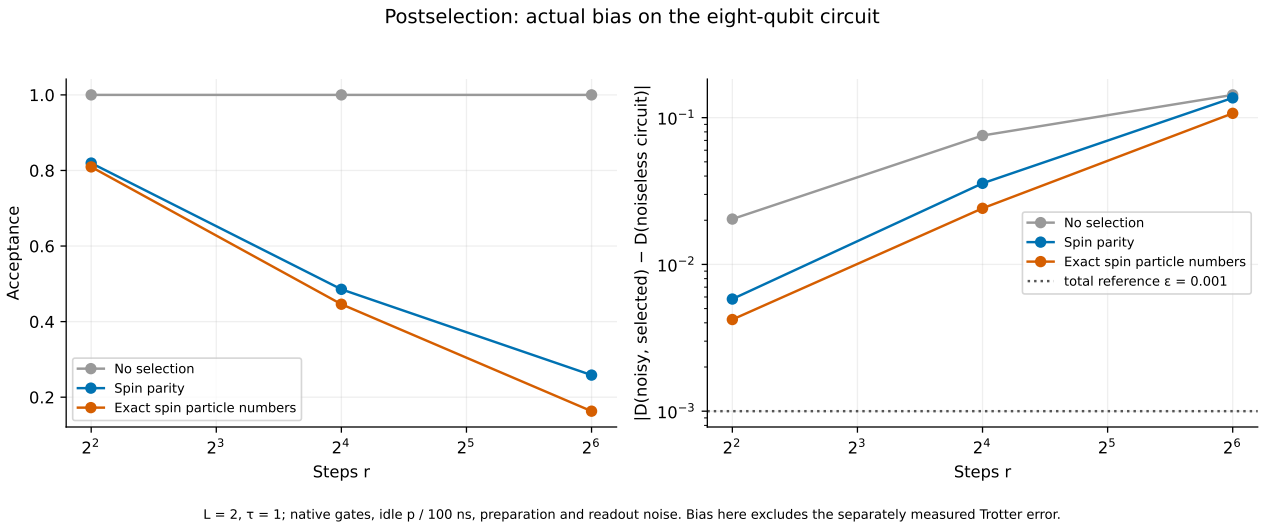


Figure 15: Exact acceptance and actual observable bias under the declared eight-qubit noise model. The right panel isolates noise bias from Trotter bias. No extrapolation from these curves to $L \geq 4$ is used.

Primitive idle channels, initial preparation flips, and final readout flips are also included. A direct Pauli-mixture implementation independently checks the optimized channel calculation. For selection event A, the measured quantity is $D_A = \mathbb{E}[D\mathbf{1}_A]/\Pr(A)$; a large or small acceptance alone says nothing sufficient about its bias.

At $r = 4$, the unselected noise bias relative to the ideal Trotter circuit is 0.020371. Selecting the two spin parities reduces it to 0.005808 with acceptance 0.81958. Selecting exact $N_\uparrow = N_\downarrow = 2$ reduces it to 0.004212 with acceptance 0.80943. Its bias relative to exact dynamics, including the independently measured Trotter error, is 0.01785. At $r = 64$, the number-selected noise bias rises to 0.1070. Postselection helps here, but by itself does not meet the strict precision target.

The parity acceptance agrees with the independently propagated Pauli-character formula. Trace errors are below 1.4×10^{-12} and the calculated density matrices remain positive. These results apply to the stated eight-qubit circuits and noise channels. They do not prove failure of selection at $L = 4$ or of combinations with other mitigation methods. The larger trapped-ion experiment [10] remains a separate circuit, geometry, and noise comparison.

15 What determines the scaling

The exposure variables give a more transferable comparison than a single crossover. With local full-lattice compilation, $R = O(Nr)$ and $\Theta = O(N\tau)$. Physical PEC and original STAR therefore have reference exponents $O(pNr)$, while the small-angle term has exponent $O(pN\tau)$. An injection floor and the plaquette fixed-angle basis changes restore $O(r)$ growth of the STAR exponent. Individual FT synthesis has $N_T = O(Nr \log(Nr/\epsilon))$ and logarithmic distance growth below threshold, but starts with a substantial protected footprint and supply overhead. These are laws of the specified models, subject to the locality and accuracy qualifications above.

The physical-error sweep in Figure 1 improves the modeled capacity of all three reference architectures as p falls. The relative ordering also depends on precision, injection floors, and source quality; holding these quantities fixed or assigning improvement laws gives different partial-FT windows. Figure 2 identifies an algorithmic change within FT that moves the strict-precision costs by approximately a factor of six. The compact-encoding scenarios and ordering diagnostic show why locality alone does

not determine either the small-system gate count or the Trotter coefficient.

The earlier compiled reference in Figure 11 retains a separate, conditional circuit comparison. At $Q = 10^6$ and strict reference precision, the largest tested conditional circuits are 2×2 for NISQ, 4×4 for STAR and FTQC, and 6×6 for the hybrid. These numbers use different implemented optimizations, conditional encoded supply, and the same unqualified $r = 4$ product formula. They are not final architecture rankings. At loose precision the encoded thresholds cluster much more closely, making delivery rate, distance, and integer replicas consequential.

Small-patch calibration explains why a common algorithmic correction does not cancel from the resource comparison: increasing r reduces a shared Trotter error but enters each architecture's cost differently. Likewise, physical truncation could reduce all three arms if its bias is controlled. Direct 4×4 checks now settle the Trotter component for the selected headline workloads. The remaining scientific dependencies are transfer of accuracy to larger lattices, a controlled spatial truncation estimate, empirical NISQ mitigation bias, and physical injection/growth and handoff schedules with residual-channel validation. Qubitization remains a separately identified later algorithm comparison.

A Matrix-free finite-lattice reference

Writing the state as a matrix indexed by up- and down-spin occupation masks, we apply

$$H\Psi = h_\uparrow \Psi + \Psi h_\downarrow^{\mathsf{T}} + V \odot \Psi. \tag{48}$$

Each hopping matrix is sparse within a single spin sector; no full-sector Hamiltonian is stored. Fermionic signs count occupied sites between the hopping endpoints in the snake order. For $U/J = 4$, the triangle bound is $\|H\| \le 2|E| + N = 64$ on 4×4. We evaluate

$$e^{-iH\tau}\psi_0 = J_0(a)\psi_0 + 2\sum_{k\ge 1}(-i)^k J_k(a) T_k(H/64)\psi_0, \qquad a = 64\tau, \tag{49}$$

using the three-term Chebyshev recurrence. The bound

$$|J_k(a)| \le \frac{(|a|/2)^k}{k!} \exp\left(\frac{a^2}{4(k+1)}\right) \tag{50}$$

follows by bounding the absolute Bessel series; successive right-hand sides have ratio at most $|a|/[2(k+1)]$. A geometric tail therefore selects a degree with state truncation error below 10^{-12}. For a $[0,1]$ observable, state error η gives observable error at most $2\eta + \eta^2$. This controls polynomial truncation, separately from floating-point error. The computed 4×4 norm error is about 10^{-13}.

Trotter hopping stages use exact two-state rotations. Plaquette stages use the fermionic Hadamard decomposition already employed by the compiler. Complete state vectors for the exact reference and both orderings agree with the independent Python implementation on 2×2, 3×2, and 3×3 to better than 2×10^{-11}. Large-run outputs retain all-site doublons, norms, input and executable hashes, and atomic case records. The step convergence criterion is a 5% agreement between the two finest observable-error coefficients, not a proof of bulk convergence.

B Logical-fault commutation and its limitation

A branch-independent fault channel $\mathcal{F}$ outside a complete PEC cancellation block, or commuting through its noise channel $\mathcal{E}$, satisfies $\mathcal{E}^{-1}\mathcal{F}\mathcal{E} = \mathcal{F}$. For a stochastic fault of probability q_L, this gives bias at most q_L for a $[0,1]$ observable. Pauli channels commute locally, making this a useful condition. It does not follow merely from linearity or branch independence.

For a counterexample, take Bloch matrices $E = \mathrm{diag}(s, s, 1)$ and $F = (1-q)I + qH$, where H is Hadamard conjugation. On $|0\rangle$, measuring $(I+X)/2$ after $E^{-1}FE$ gives bias $q/(2s) > q$ when $s < 1/2$. This is a stochastic non-Pauli fault inside the cancellation sandwich. The logical STAR instrument of Section 12.1 instead verifies local Z covariance under its stated Pauli fault model. A physical implementation must justify that model across injection, growth, retries, and any selection before the more favorable budget becomes its default.

C Compilation, checks, and reproducibility

The installed CPU CUDA-Q Logical environment compiles the common analog program, native physical circuits, and synthesized Clifford+T words. The physical routes carry mode permutations, explicit nearest-neighbor gates, ASAP schedules, counts, and hashes. The STAR retry program contains real logical preparation, measurement, and conditional angle doubling with an exhaustion flag. These P0 artifacts do not imply a physical STAR QEC schedule.

The installed minimal surface recipe lacks some needed gate gadgets. An explicit encoded macro adapter reaches physical IR, but the Hubbard backend timing is rejected. A reduced circuit with four fixed-angle rotations repeated four times, totaling 16 T states, triggers a nested schedule interval error. The 4-, 8-, and 32-T variants pass a loose bound, and separate two-resource template-call controls return their expected times. The root cause remains unresolved. None of these backend timings enters the resource curves; passing the loose controls is not a validation of physical encoded timing.

Tests independently check fermionic signs, product-formula identities, synthesis accuracy, exact-sector dynamics, primitive Pauli noise, STAR instrument algebra, and resource budgets. The capacity inversion additionally checks threshold inclusion, nonmonotone cost envelopes, and the untested region. These checks validate the specified calculations rather than the hardware assumptions.

The repository entry point is `README.md`. Source code is in `python/`, tests in `tests/`, assumptions in `configs/`, and this manuscript in `tex/main.tex` with a dedicated `tex/references.bib`. Current overview figures are in `figures/`; versioned scientific evidence stays in `results/`. Earlier complete notes are in `tex/archive/`. The original v3 manifest is preserved: historical source hashes are checked against their archived snapshots, while the relocated code replays the same saved numerical evidence. The presentation manifest records the new figures and their source datasets without relabeling older results as a new calculation.

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